Trust-Gated Capability Control: An Operational Framework for Breaking the Trust–Vulnerability Paradox in Multi-Agent LLM Systems

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Abstract

Layered trust models for multi-agent large language model (LLM) systems remain largely conceptual: they describe which dimensions of trust matter but do not specify how trust at different layers combines, how layer importance adapts at runtime, or how trust should govern what an agent is permitted to do. This gap is consequential because increasing inter-agent trust raises task success while simultaneously enlarging exposure to adversarial exploitation, a tension formalized as the Trust-Vulnerability Paradox (TVP). We make a five-layer trust stack operational. We first introduce a cross-layer synergy operator that encodes layer dependencies through a lower-triangular coupling matrix, and a generalized power-mean composite trust that admits a closed-form conservative fixed point and provable bounds. We then prove that failure-grounded multiplicative-weights estimation of layer importance converges geometrically and is no-regret. Building on these, we introduce Trust-Gated Capability Control (TGCC): a runtime mechanism that issues short-lived, revocable capability grants conditioned on both composite trust and per-prerequisite-layer thresholds. We prove that TGCC breaks the paradox, that its revocation latency is logarithmic, and that the false-positive rate for honest agents decays exponentially with interaction length. A deployment-time threshold-calibration procedure with a sample-complexity guarantee connects the theory to practice. Five experiments validate the framework: a real-LLM epistemic calibration probe shows the synergy operator reduces over-exposure by a factor of 4.2 against a no-synergy baseline; a multi-agent debate study confirms complete sleeper revocation across team sizes; an ablation identifies the practical operating region; a code-generation pipeline blocks 95% of harmful commits; and a scalability study confirms constant per-agent overhead through fifty agents, empirically validating the quadratic complexity bound. An extended robustness study of eleven further experiments (Appendix H) addresses statistical significance, colluding adversaries, indirect prompt injection, per-layer attribution, long-horizon drift, parameter sensitivity, revocation-then-recovery, Byzantine chains, auditability, calibration robustness, and adaptive attackers.

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CCS Concepts

• **Security and privacy** → **Systems security**; *Access control*; • **Computing methodologies** → *Multi-agent systems*; Online learning settings.

Keywords

Multi-agent systems, large language models, trust management, zero-trust architecture, capability-based security, agent safety, alignment, calibration, online learning, power-mean aggregation, threshold calibration, adversarial robustness

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1 Introduction

Large language model (LLM) agents are no longer deployed only as solitary assistants. They are increasingly composed into *collectives* that plan, negotiate, delegate, and act through shared tools, memories, and message channels [22, 24, 54, 56]. Such collectives already drive software engineering pipelines, scientific-discovery workflows, clinical-decision support, and autonomous web agents. As autonomy and interconnection grow, the behaviour of any one agent becomes consequential for the entire system: an output that is consumed downstream without scrutiny can silently propagate errors, and a single subverted participant can compromise a whole workflow. Collaboration, in other words, converts *trust* between agents from a background assumption into a first-class operational variable that must be measured, reasoned about, and acted upon at runtime.

Recent empirical work has made this concern precise. Structured attacks that exploit inter-agent trust raise harmful-response rates well above single-agent baselines [46], and a growing body of evidence shows that trust and vulnerability are coupled rather than independent. Specifically, raising the degree to which agents accept one another's claims improves task success but simultaneously enlarges the attack surface, a phenomenon recently formalized as the *Trust-Vulnerability Paradox* (TVP) [59]. The lesson echoes the zero-trust movement in systems security: trust should never be implicit, should be continuously re-verified, and should be coupled to least-privilege authorization [39, 49, 51]. Yet translating that principle into a concrete, runtime decision rule for LLM collectives remains an open problem.

The conceptual–operational gap. A natural and increasingly popular response is to model trustworthiness as a *layered stack*, decomposing it into dimensions such as epistemic reliability, behavioural alignment, role adherence, social accountability, and institutional governance. Layered models are attractive because they separate concerns and map onto distinct measurement instruments. However, the existing accounts are almost entirely *descriptive*. They tell us *which* dimensions matter, but they leave three operational questions unanswered. First, *composition*: given that the layers are not independent—epistemic failure undermines every higher layer—how should the layer-wise trust values be combined into a single, actionable score? Second, *weighting*: how is the relative importance of each layer set, and can that importance adapt to whichever layer is currently failing? Third, *actuation*: once a composite trust score exists, how should it govern what an agent is permitted to do, and how quickly can permissions be withdrawn? Without principled answers to all three, a layered model is a taxonomy, not a controller, and cannot be deployed in a safety-critical setting.

A motivating scenario. Consider a clinical pipeline ending in a *scribe* agent that holds a high-risk *write-to-EHR* capability. Suppose an upstream *reasoner* is compromised so that it returns plausible-sounding but incorrect inferences with high stated confidence, while remaining cooperative and articulate in tone. A controller that authorizes the write based on apparent helpfulness sees nothing wrong—fluent, well-formatted, and professional—so incorrect conclusions are committed to patient records. This is the TVP in microcosm: behavioural agreeableness masks a collapse of epistemic reliability, and a trust signal that conflates the two is exploited. The defence we seek must detect that the *prerequisite* layer has failed even though the *visible* layer appears healthy, and withdraw the capability before harm is done. We return to this scenario in Appendix F once TGCC is fully defined.

Why existing solutions fall short. Scalar reputation systems cannot represent a layer-specific failure hidden behind cooperative behaviour [27, 48]; zero-trust policies advocate per-call revocation but give no decision rule for *when* to revoke on multi-dimensional evidence [39, 51]; the defences proposed alongside the TVP operate *around* the trust signal rather than reshaping how it is computed [59]; and training-time alignment [3, 9, 41] shapes dispositions but provides no runtime mechanism against an agent that turns adversarial after deployment. None of these lines provides a compositional, runtime-operational trust framework with provable guarantees.

Our approach. We treat the layered trust stack as a *dynamical object* made operational through four connected components: a *cross-layer synergy operator* that propagates a prerequisite-layer deficit upward and discounts every layer that depends on it; a power-mean *composite* that aggregates the effective per-layer trusts into a single score dominated by the weakest verified layer; a failure-grounded *adaptive-weighting* rule that re-routes the composite's sensitivity toward whichever layer is currently most responsible for harm; and *Trust-Gated Capability Control* (TGCC), which turns the composite into short-lived, revocable capability grants so that authorization tracks the weakest verified layer rather than the easily inflated behavioural layer. We prove that this construction breaks the TVP, bounds the time to contain a compromise, and keeps false positives for honest agents exponentially rare.

Contributions.

- We formalize the multi-agent trust-gating problem and the Trust-Vulnerability Paradox as a precise property of authorization policies (Section 3).
- We introduce a cross-layer synergy operator and a generalized power-mean composite trust, and establish a closed-form conservative fixed point, tight bounds, a weakest-link limit, monotonicity, and sensitivity properties (Section 4).
- We ground per-layer importance weights in observed failures and prove that the resulting estimator is no-regret and converges geometrically (Section 5).
- We present the TGCC algorithm and prove that it has constant per-step overhead, that it breaks the paradox, that its revocation latency is logarithmic, and that the false-positive rate decays exponentially (Section 6).
- We give a deployment-time threshold-calibration procedure with correctness and sample-complexity guarantees (Section 7), and additional adversarial-robustness results (Section 8).
- We validate the framework with five experiments on real LLMs and simulation (Section 9), and provide a step-by-step deployment protocol (Section 10).
- We complement the main study with a compact summary (Section 9.6) and eleven further experiments (Appendix H) covering statistical significance, colluding adversaries, prompt injection, per-layer attribution, long-horizon drift, parameter sensitivity, recovery, Byzantine chains, audit trails, calibration robustness, and adaptive attackers.

2 Related Work

Multi-agent LLM systems. Systems such as AutoGen [58], CAMEL [36], MetaGPT [24], ChatDev [47], and AgentVerse [8] orchestrate role-specialized agents through structured dialogue, while embodied agents [43, 55] extend the paradigm to open-ended environments; surveys [22, 54, 56] document a common architecture of roles, messages, and tools. These works focus on *capability*, whereas we focus on *control*: bounding what a collective is permitted to do as a function of measured trust. Our framework is complementary, protocol-agnostic, and can wrap any of these systems.

Trust as a security variable. The closest motivation for our work is the formalization of the TVP [59], which isolates three facets of inter-agent trust—strength, scope, and revocability—and shows that increasing *scalar* trust monotonically raises both over-exposure and authorization drift. Xu et al.'s defences accept scalar trust as given and reduce exposure by tuning *around* it (thresholding, rate-limiting) rather than reshaping how the signal is computed. TGCC differs on three axes: (i) we lift trust from a scalar to a coupled layered vector, so an epistemically broken but behaviourally polished agent is legible at all; (ii) we prove the paradox is broken (Theorem 3) at the operator level, not merely observed empirically; and (iii) we ship a runtime capability gate with logarithmic revocation latency (Corollary 1) and exponentially decaying honest false positives (Theorem 4). Zero-trust treatments for agentic AI [39, 51] and TRiSM [49] advocate verifiable identity and continuous verification but stop short of a concrete multi-layer decision rule; we supply exactly the missing rule.

Reputation and probabilistic trust. Trust and reputation have a long history in multi-agent systems [48, 52], including EigenTrust [28], the Beta reputation system [27], and subjective logic [26]; LLM-era revivals include DynaTrust [37], AgentGuard [30], and economic-game trust elicitation [61]. These methods track a *single* trust quantity and are therefore blind to the core TVP scenario, in which an agent is behaviourally cooperative yet epistemically broken. We retain the mathematically convenient Beta machinery and prove a conservative fixed point for it, but lift trust from a scalar to a coupled *vector*, which is what makes a hidden prerequisite failure visible. Input/output content-filter systems such as LLM-Guard [45] and NeMo Guardrails [50] operate on a different axis (per-message text screening) and are complementary to any runtime trust controller; TGCC composes with them rather than replacing them.

Debate, deception, and collective failure. Multi-agent debate can improve factuality [12, 38], but the same channels also enable harm: structured jailbreaks injected into one debater amplify harmful responses [46], and social deception simulations show agents readily form, exploit, and misjudge trust [10]. Sleeper-agent behaviour [25] shows models can behave benignly until a trigger condition is met—exactly the threat profile our controller contains.

Uncertainty and calibration. The epistemic layer rests on knowing when an agent is wrong. Neural networks are frequently miscalibrated [21], and predictive uncertainty can be quantified via Bayesian approximations [15], deep ensembles [33], semantic entropy [32], and selective prediction [18]; related work propagates epistemic state across collaborating agents [63]. Our synergy operator converts such a passive calibration measurement into an active control input that gates downstream capabilities.

Prompt injection, tool use, and capability security. Indirect prompt-injection [20] and jailbreaks [57] subvert an agent through its inputs, while tool-use frameworks [53, 62] expand the actions an agent can take; TGCC gates those actions on composite trust rather than detecting every input exploit. Governing a principal via unforgeable, least-privilege tokens dates to capability-based operating systems [11] and the object-capability model [40], and underlies zero-trust architecture [51]; our grant rule is precisely such a capability-issuance policy, short-lived and revoked the moment trust degrades.

Online learning, aggregation, and Byzantine fault tolerance. Our adaptive weighting is an instance of the multiplicative-weights / Hedge family [2, 7, 14], and our composite trust is a weighted power mean [23] related to ordered weighted averaging [60] (Appendix C justifies this choice). Byzantine fault tolerance [6, 34] gives correctness guarantees for a bounded fraction of arbitrary faults but assumes a $3f + 1$ replication factor impractical for LLM agents; formal verification [17, 29] and temporal monitoring [31] likewise do not scale to the interaction space of LLM collectives, so we trade hard certificates for probabilistic bounds. Our Beta estimator with a forgetting factor is, in effect, a bank of online change-point detectors [4, 42], one per layer.

Alignment, human–AI trust, and governance. RLHF [9, 41], constitutional methods [3], red-teaming [44], and governance-graph collusion control [5] address specific trust layers; since deployed agents face non-stationary environments [16], our forgetting factor keeps trust responsive to drift, and human–automation calibration [35] motivates our conservative fixed point (an over-eager controller erodes honest agents just as over-trust endangers a human operator). These works populate individual layers of our stack; we contribute the operator that composes them and the controller that acts on the composite. Table 3 in Appendix A positions our approach against the most comparable lines of work.

3 System Model and Problem Formulation

3.1 Multi-Agent LLM System

A multi-agent LLM system is a tuple $\mathcal{S} = (\mathbb{A}, C, \mathcal{P}, \mathcal{K})$: $\mathbb{A} = \{\mathcal{A}_1, \ldots, \mathcal{A}_N\}$ is a finite set of agents, each backed by an LLM; C is a finite set of *capabilities* such as tool calls, message scopes, and write actions; $\mathcal{P}$ is the capability-assignment policy the system must enforce; and $\mathcal{K}$ is a shared knowledge store agents read from and write to subject to $\mathcal{P}$. Agents interact over a directed *message graph* in which an edge from agent i to agent j means i's outputs are consumed by j without independent re-verification. A *controller* observes per-agent check signals and issues or revokes capability tokens; it is the object we design, everything else is given.

3.2 Trust as a Multi-Faceted Security Variable

Following the decomposition that motivates the TVP [59], inter-agent trust has three components: *strength* (how readily one agent accepts another's claims without verification), *scope* (which capabilities are authorized on that basis), and *revocability* (how quickly permissions can be withdrawn after a violation). The paradox arises because conventional policies make scope a monotone function of strength; since strength is easy to inflate by behaving cooperatively while a hidden capability such as epistemic reliability degrades, such policies conflate behavioural agreeableness with operational trustworthiness. The design goal is to make scope depend not on raw strength but on *verified* trust across all relevant layers.

3.3 Threat Model

We adopt a strong but realistic adversary that captures the stealth behaviour observed in sleeper-agent studies [10, 25].

Assumption 1 (Stealth sleeper adversary)**.** A compromised agent (i) behaves cooperatively for an initial dormant period in order to accrue trust, (ii) begins, at an activation time, to inject incorrect or harmful outputs while maintaining a cooperative surface form, and (iii) can inflate the behavioural-layer signal arbitrarily while degrading the epistemic-layer signal. The adversary controls one agent and has full knowledge of the controller's parameters but cannot forge the check signals of layers it does not control.

This adversary is deliberately stronger than a naive faulty node: it is strategic, it knows the defence, and it specifically targets the gap between behaviour and epistemics that scalar trust systems cannot see. Appendix A situates it within a broader taxonomy of adversaries and states which result addresses each.

3.4 Problem Statement

A controller is vulnerable to the paradox if an adversary can *simultaneously* appear more trustworthy and gain more privilege.

Definition 1 (Trust-Vulnerability Paradox)**.** A controller suffers the *Trust-Vulnerability Paradox* if there exists an adversary that simultaneously (a) increases the trust strength the controller observes and (b) gains access to a strictly larger capability scope, by degrading an orthogonal trust dimension while maintaining or raising the dimension the controller monitors.

Our objective is to design a controller that is *free* of the paradox in the sense of Definition 1, that is computationally efficient enough to run on every interaction, and that does not unduly penalize honest agents. The remainder of the paper constructs such a controller and proves these three properties in turn.

4 The Operational Trust Stack

> ***Core idea.*** *Maintain a* vector *of per-layer trusts, not a single scalar; couple layers so a prerequisite deficit propagates upward and suppresses dependants; aggregate into a composite dominated by the weakest verified layer; and gate every capability on that composite, so authorization tracks the weakest verified layer rather than the behavioural trust an adversary can inflate.*

4.1 Per-Layer Trust as a Beta Belief

We use five trust layers, ordered by dependency: epistemic, behavioural, role, social, and institutional. For each agent we maintain a trust vector whose components are the means of per-layer Beta posteriors, updated online from check signals with two deliberate design choices borrowed from safety-critical reputation modelling [27, 37]: a forgetting factor that keeps trust responsive to behavioural shifts and prevents an agent from "banking" a long history of good behaviour, and an asymmetric penalty that makes trust fall faster than it rises. Writing $s_\ell(t) \in [0,1]$ for the layer-ℓ check signal, the pseudo-count update is

$$a_\ell \leftarrow \gamma a_\ell + s_\ell, \qquad b_\ell \leftarrow \gamma b_\ell + \omega(1 - s_\ell), \tag{1}$$

with forgetting factor $\gamma \in (0,1)$ and failure penalty $\omega \geq 1$, and the layer trust is the posterior mean $T_\ell = a_\ell/(a_\ell + b_\ell)$. The asymmetry $\omega > 1$ encodes a conservative stance: a single failure costs more than a single success earns.

Proposition 1 (Conservative trust fixed point)**.** *Suppose layer ℓ succeeds with stationary probability ρ_ℓ. Then the expected-evidence recursion induced by* (1) *has the unique fixed point*

$$T_\ell^\star = \frac{\rho_\ell}{\rho_\ell + \omega(1 - \rho_\ell)}, \tag{2}$$

which satisfies $T_\ell^\star \leq \rho_\ell$ with equality iff $\omega = 1$, is strictly increasing in ρ_ℓ, and is approached geometrically at rate γ (proof in Appendix G).

Remark 1. For a reliable layer with $\rho = 0.9$ and $\omega = 3$, the fixed point is 0.75; for a collapsed layer with $\rho = 0.3$, it is approximately 0.13. A modest drop in reliability is thus amplified into a sharp, conservative drop in trust.

4.2 Cross-Layer Synergy Operator

The five layers are not independent: epistemic reliability underpins behavioural alignment, which in turn bounds role adherence, and so on up the stack. A purely per-layer view would miss the fact that an epistemic collapse should drag down our confidence in every dependent layer, even if those layers' own signals look healthy. The dependencies are encoded by a lower-triangular *coupling matrix* whose entry $C_{\ell k}$ (for $k < \ell$) is the degree to which layer ℓ depends on prerequisite k (a concrete default coupling matrix and a worked numeric example appear in Appendix B).

Definition 2 (Synergy operator)**.** The *effective trust* of layer ℓ is its own trust discounted by the deficits of its prerequisites,

$$\tilde{T}_\ell = T_\ell \prod_{k<\ell} \big(1 - C_{\ell k}(1 - T_k)\big). \tag{3}$$

A deficit $1 - T_k$ in prerequisite k reduces the effective trust of layer ℓ by a factor that scales with the coupling strength; zero coupling leaves the layer independent, and full coupling with a fully failed prerequisite zeroes the dependent layer's effective trust regardless of its own signal.

Proposition 2 (Synergy monotonicity)**.** *For each prerequisite $k < \ell$, the effective trust $\tilde{T}_\ell$ is non-decreasing in T_k, with non-negative partial derivative*

$$\frac{\partial \tilde{T}_\ell}{\partial T_k} = C_{\ell k}\, T_\ell \prod_{k'<\ell,\, k' \neq k} \big(1 - C_{\ell k'}(1 - T_{k'})\big) \geq 0. \tag{4}$$

Proof. Differentiate (3) with respect to T_k; the surviving factor is $C_{\ell k}$ times T_ℓ times the product over the remaining prerequisites, each non-negative. □

4.3 Composite Trust and Its Bounds

The controller must reduce the effective per-layer trusts to a single number on which to gate. A weighted power mean of negative order lets a single weak layer veto a high grant while still distinguishing one marginal layer from several (Appendix C compares this choice against the alternatives). Writing $\boldsymbol{\alpha}$ for adaptive weights on the probability simplex, the composite is

$$\Phi_p(\mathbf{T}; \boldsymbol{\alpha}) = \Big(\textstyle\sum_{\ell=1}^{L} \alpha_\ell\, \tilde{T}_\ell^{\,p}\Big)^{1/p}, \qquad p < 0. \tag{5}$$

As $p \to -\infty$ the composite is pulled toward the minimum effective trust, recovering the conservative weakest-link rule; for moderate $|p|$ it permits limited compensation among healthy layers.

Lemma 1 (Well-posedness)**.** *For trust values in the unit interval, a lower-triangular coupling matrix with entries in the unit interval, weights on the simplex, and any $p < 0$, the effective trusts lie in $[0,1]$ and the composite lies between the minimum and maximum effective trust, hence in $[0,1]$.*

Theorem 1 (Bounds and weakest-link limit)**.** *For any $p < 0$, the composite Φ_p is non-decreasing in each effective trust and is sandwiched between the minimum effective trust and the weighted arithmetic mean,*

$$\min_\ell \tilde{T}_\ell \;\leq\; \Phi_p(\mathbf{T}; \boldsymbol{\alpha}) \;\leq\; \sum_\ell \alpha_\ell \tilde{T}_\ell, \tag{6}$$

with the lower bound attained as $p \to -\infty$. Moreover, the composite is pinned to the weakest layer up to a bounded multiplicative slack,

$$\Phi_p(\mathbf{T}; \boldsymbol{\alpha}) \;\leq\; \kappa \min_\ell \tilde{T}_\ell, \qquad \kappa := \big(\min_\ell \alpha_\ell\big)^{1/p} \geq 1. \tag{7}$$

Proof. The bounds and the limit are the standard ordering of weighted power means [23]. For the slack bound, let j index the smallest effective trust. Since $p < 0$, $\sum_\ell \alpha_\ell \tilde{T}_\ell^p \geq \alpha_j \tilde{T}_j^p$; applying the decreasing map $x \mapsto x^{1/p}$ and using $\alpha_j \geq \min_\ell \alpha_\ell$ yields the bound. □

The slack factor κ is the precise lever the controller will use: because the composite can never exceed κ times the weakest layer, driving any single prerequisite layer toward zero drives the composite toward zero as well.

Lemma 2 (Trust sensitivity). *The partial derivative of the composite with respect to effective trust $\tilde{T}_\ell$ is positive and inversely related to that layer's trust,*

$$\frac{\partial \Phi_p}{\partial \tilde{T}_\ell} = \alpha_\ell \, \tilde{T}_\ell^{\,p-1} \Big(\sum_k \alpha_k \tilde{T}_k^{\,p} \Big)^{(1/p)-1} > 0. \tag{8}$$

Because $p - 1 < 0$, the factor $\tilde{T}_\ell^{\,p-1}$ grows without bound as $\tilde{T}_\ell \to 0$, so the composite is disproportionately sensitive to the weakest layer.

Proposition 3 (Cascade containment). *If a prerequisite layer k is compromised so that $T_k \leq \delta$, then every dependent layer $\ell > k$ with coupling at least c satisfies $\tilde{T}_\ell \leq 1 - c(1 - \delta)$. In particular, if the epistemic layer is a prerequisite of all layers, then the composite obeys $\Phi_p \leq \kappa\delta$ (proof in Appendix G).*

4.4 Collective Trust in Multi-Agent Topologies

The preceding results concern a single agent. In a collective, capabilities are often exercised at the end of a chain of agents, and trust composes *across* agents by another weakest-link principle.

Theorem 2 (Collective trust bound). *Consider N agents arranged in a pipeline, where each agent's outputs are consumed by the next without filtering, and let each agent's individual composite trust be $\Phi_p^{(i)}$. A collective capability that requires the whole chain is granted only if every agent is individually granted, so the effective collective composite trust is at most the minimum individual composite, $\Phi_p^{\text{collective}} \leq \min_i \Phi_p^{(i)}$. For a general directed acyclic dependency graph, the bound holds along every root-to-leaf path (proof in Appendix G).*

Remark 2. Theorem 2 means the controller does not need global coordination: to contain a compromise it suffices to revoke a single agent in the chain.

4.5 Layer Signals and Notation

The framework is agnostic to how each check signal is produced; concrete probe implementations for the five canonical layers, and a table of the notation used throughout, are given in Appendix D. For text agents the epistemic signal, derived from calibration error, is the most discriminating because it is precisely the quantity a stealth adversary cannot inflate while remaining wrong.

5 Failure-Grounded Adaptive Weighting

The weights in the composite determine which layers it attends to most. Fixing them in advance is brittle: the layer that matters most is the one currently failing, and that changes over time and across deployments. The controller maintains, for each layer, an exponentially weighted estimate of the rate at which that layer is the attributed root cause of harm, and turns those estimates into weights through a soft-max—the Hedge, or multiplicative-weights, update [2, 14]. Writing $q_\ell(t)$ for the layer-attributed loss observed at step t, the estimate and weights evolve as

$$\hat{r}_\ell(t) = (1 - \lambda)\hat{r}_\ell(t-1) + \lambda \, q_\ell(t), \tag{9}$$

$$\alpha_\ell(t) = \frac{\exp(\eta \, \hat{r}_\ell(t))}{\sum_k \exp(\eta \, \hat{r}_k(t))}. \tag{10}$$

Because the weights concentrate on the layer with the highest attributed loss, the composite becomes most sensitive precisely where the system is breaking, complementing Lemma 2. Appendix E (Fig. 4b) shows this behaviour in simulation: when the epistemic layer begins to fail, weight migrates toward it within a few steps.

Lemma 3 (No-regret layer tracking). *With learning rate $\eta = \sqrt{8 \ln L / T}$, the weighting* (10) *incurs cumulative regret at most $\sqrt{(T/2)\ln L}$ relative to the best fixed weighting in hindsight over horizon T.*

Proof. This is the standard regret bound for Hedge with losses bounded in the unit interval [7, 14]. □

Proposition 4 (Adaptive-weight convergence). *Under stationary layer-attributed loss rates, the estimates* (9) *converge geometrically at rate $1 - \lambda$ to those rates, and the weights* (10) *converge at the same rate to the soft-max of the scaled rates.*

Proof. The estimate recursion is a contractive affine map whose unique fixed point is the loss rate, with contraction factor $1 - \lambda$. Since the soft-max is smooth, convergence of the estimates implies convergence of the weights at the same geometric rate. □

6 Trust-Gated Capability Control

6.1 The Grant Rule

With a composite trust in hand, the controller must decide what each agent may do. We attach to every capability a composite risk threshold and a set of prerequisite layers, each with its own threshold; a capability is granted only when the composite clears its threshold *and* every prerequisite layer clears its own.

Definition 3 (Trust-gated grant). An agent is granted capability c at step t exactly when its composite trust is at least the composite threshold and each required prerequisite layer's effective trust is at least its per-layer threshold,

$$g(c, t) = \mathbb{1}\Big[\Phi_p \geq \theta_c \ \wedge \ \forall \ell \in \mathrm{req}(c) : \ \tilde{T}_\ell \geq \theta_{c,\ell}\Big]. \tag{11}$$

Because the grant is recomputed at every step, a capability token is inherently short-lived: it persists only as long as trust supports it and is withdrawn automatically the moment trust degrades, realizing least-privilege and revocability without an explicit revocation message [40, 51]. Algorithm 1 states the full runtime loop.

Proposition 5 (Constant per-step overhead). *Each iteration of Algorithm 1 runs in time $O(L^2 + |C|)$ and uses $O(L)$ memory, independent of the interaction-history length (proof in Appendix G).*

Algorithm 1 TGCC Runtime Loop (per agent, per step)

Require: coupling $\mathbf{C}$, order p, thresholds $\{\theta_c, \theta_{c,\ell}\}$, rates λ, η, recovery threshold θ_{rec}
observe layer checks $s_1, \ldots, s_L$; update Beta beliefs via (1)
compute effective trusts $\tilde{\mathbf{T}}$ via the synergy operator (3)
update loss estimates and weights $\boldsymbol{\alpha}$ via (10)
compute composite Φ_p from (5)
for all capabilities c **do**
  issue or keep the token for c if the grant rule (11) holds; else revoke
end for
if $\min_\ell \tilde{T}_\ell < \theta_{\text{rec}}$ **then**
  trigger layer-specific recovery (e.g. external knowledge anchoring)
end if

6.2 Breaking the Paradox

The adversary of Assumption 1 can raise the behavioural layer as much as it likes, but the grant rule conditions on the composite, which is pinned to the weakest verified layer by Theorem 1 and Proposition 3.

Theorem 3 (Paradox resolution). *Consider the stealth adversary of Assumption 1, who drives the behavioural trust arbitrarily high while degrading a prerequisite layer to at most δ, where that layer is a prerequisite of all layers. Under naive behavioural gating every capability whose threshold is at most the behavioural trust remains granted, so over-exposure is unbounded in the depth of the compromise. Under TGCC, every capability whose threshold exceeds $\kappa\delta$ is revoked irrespective of the behavioural trust, and the granted high-risk set is non-increasing in the compromise depth.*

Proof. By Proposition 3 the compromise forces the composite to at most $\kappa\delta$, so any capability with threshold above $\kappa\delta$ fails the first conjunct of the grant rule and is denied, independently of the behavioural layer. A deeper compromise (smaller δ) only lowers the bound and hence weakly enlarges the revoked set. The naive case is immediate because its grant rule depends only on the behavioural trust. □

Corollary 1 (Self-revocation with bounded latency). *After a prerequisite layer's reliability drops to a fixed point at most δ, the layer's trust converges geometrically and, once it falls below δ, every capability with threshold above $\kappa\delta$ is revoked. The number of steps to reach an ϵ-neighborhood of the fixed point is $O(\log_{1/\gamma}(1/\epsilon))$.*

Figure 2 in Appendix E illustrates this dynamic in simulation: the behavioural layer stays high, exactly as the adversary intends, yet the epistemic layer collapses and drags the composite below threshold within a few steps.

6.3 Protecting Honest Agents

A controller that revokes aggressively is only useful if it does not also punish honest agents. The key quantity is the effective sample count, which grows toward a finite ceiling set by the forgetting factor.

Theorem 4 (False-positive bound). *Consider an honest agent whose per-layer reliabilities are bounded below, so that its composite trust at the fixed point strictly exceeds the composite threshold and each prerequisite layer's fixed point strictly exceeds its per-layer threshold. Let the effective sample count at step t be $n_\ell(t) = (1-\gamma^t)/(1-\gamma)$. Then the probability that the honest agent is wrongly denied a capability is at most*

$$L \exp\!\big(-2(T_\ell^\star - \theta_{c,\ell})^2\, n_\ell(t)\big), \tag{12}$$

which decays to zero exponentially as t grows.

Proof. Each layer trust is a posterior mean concentrating around its fixed point with effective count $n_\ell(t)$. By Hoeffding's inequality the probability that a layer trust falls below its threshold is at most $\exp(-2(T_\ell^\star - \theta_{c,\ell})^2 n_\ell(t))$. A union bound over the prerequisite layers (the composite check is implied because the honest composite exceeds its threshold by assumption) gives (12). □

Key insight. TGCC makes authorization increasing in the weakest cross-layer-verified trust rather than raw behavioural trust, so the paradox's unbounded exposure growth becomes bounded, self-revoking exposure with logarithmic detection latency, while honest agents face exponentially decaying false-positive risk.

7 Threshold Calibration

A guarantee is only as useful as the thresholds it is instantiated with. A central empirical finding of Section 9 is that composite trust occupies a markedly lower numerical range when driven by real LLM signals than in idealized simulation, because the synergy operator discounts higher layers under limited, noisy evidence. A threshold transplanted from simulation would therefore be miscalibrated, so we treat threshold selection as a data-driven step.

Algorithm 2 (given in full in Appendix E) runs the controller in observe-only mode over a short pilot of honest interactions, records the composite and per-layer trusts after burn-in, and sets each threshold a safety margin below the observed honest minimum: the composite threshold to the observed minimum composite minus ϵ, and each prerequisite threshold to the observed minimum layer trust minus ϵ, verifying that at least a $1-\delta$ fraction of pilot steps would have been admitted (shrinking ϵ otherwise).

Proposition 6 (Calibration correctness). *Let the gap between the honest fixed-point composite and the observed pilot minimum be Δ. With safety margin smaller than $\Delta/2$, Algorithm 2 returns a composite threshold that is below the honest fixed point and above the fixed point minus Δ, so it neither chronically denies honest agents nor misses reliability drops of magnitude at least $\Delta/2$.*

Theorem 5 (Calibration sample complexity). *To estimate the honest minimum composite to accuracy $\epsilon/2$ with probability at least $1-\delta$, it suffices to observe a pilot of length*

$$N_{\text{pilot}} \;\geq\; \frac{1}{1-\gamma} + \frac{1}{2\epsilon^2} \ln \frac{2}{\delta}, \tag{13}$$

where the first term is the burn-in and the second the post-burn-in sample requirement.

Proof. After burn-in the observations are approximately independent with bounded variance. The sample requirement for an

$\epsilon/2$-accurate empirical minimum at confidence $1-\delta$ follows from Hoeffding's inequality; adding the burn-in gives (13). □

Remark 3. For $\gamma = 0.985$ with accuracy and confidence parameters of 0.05, the bound calls for roughly three hundred pilot steps. In our real-LLM experiments the burn-in is about thirteen steps and twenty pilot questions suffice at a looser confidence level, so calibration is cheap in practice. Calibration fixes only the numerical thresholds; the paradox-resolution, latency, and false-positive guarantees (Thms. 3, 4; Cor. 1) hold irrespective of the specific threshold values chosen.

8 Adversarial Robustness Analysis

The paradox theorem addresses the canonical stealth adversary. We strengthen the analysis along two further axes: an attacker confined to a single layer, and a fundamental limit on detection speed.

Corollary 2 (Single-layer attack impossibility)**.** *Suppose a prerequisite layer is required by every high-risk capability and is coupled to all dependent layers with strength at least $c_{\min}$. Then an adversary who controls only that layer's signal, with all other signals honest, cannot escalate to any capability whose threshold exceeds $\kappa(1-c_{\min})$, regardless of the behavioural layer.*

Proof. Controlling one layer's signal cannot raise the others. By cascade containment the composite is at most κ times the controlled layer's trust; even at the most favourable value the adversary can sustain, the composite stays below the threshold by assumption. □

Theorem 6 (Detection-latency lower bound)**.** *After the compromise, the number of steps before the composite first falls below the threshold is at least*

$$\frac{1}{1-\gamma}\ \ln\frac{T_k(t_0)-\theta_c/\kappa}{T_k^\star-\theta_c/\kappa}, \tag{14}$$

where $T_k(t_0)$ is the layer trust at the moment of compromise and $T_k^\star$ its post-compromise fixed point.

Proof. The layer trust converges geometrically at rate γ from its pre-compromise value to its post-compromise fixed point. Detection requires it to fall below θ_c/κ by cascade containment; solving the geometric trajectory for the crossing time gives (14). □

Remark 4. Smaller γ detects faster but is jumpier under transient noise; larger γ is smoother but slower. An adversary that calibrates its stated confidence to its reduced accuracy keeps the epistemic signal high but pays a persuasion cost: E16 (Appendix H.11) verifies that perfect calibration raises OER only from 0.17 to ≈ 0.22 while attacker confidence collapses to $\rho \approx 0.30$, well below the persuasion threshold of any high-risk capability, so the threat reduces to low-quality output the role and social layers are designed to catch.

9 Experimental Evaluation

We evaluate the framework with five experiments spanning real LLMs and controlled simulation. The real-LLM experiments test whether the predicted phenomena—the paradox, its resolution, and the role of the synergy operator—actually arise with genuine models, while the simulations isolate the contribution of individual components and probe scalability.

9.1 Protocol, Baselines, and Metrics

Unless stated otherwise, all experiments use the default coupling matrix of Appendix B, forgetting factor 0.985, failure penalty 3, power-mean order −6, and Hedge learning rate 0.5 (sensitivity swept in Experiment 3, tabulated in Appendix E). Real-LLM experiments use two production LLMs (Anthropic Claude Sonnet 4.5 and Azure OpenAI `gpt-5`) through their APIs with per-layer checks derived from LLM probes; cross-provider bootstrap CIs are reported in Appendix H.1. Simulations draw Bernoulli check signals from latent reliabilities. We compare three conditions throughout: the full controller (TGCC); an ablation removing the synergy coupling (TGCC-NoSyn); and a naive baseline authorizing on the behavioural layer alone (Naive). The primary metrics are the over-exposure rate (post-compromise steps granting a high-risk capability), the revocation latency (steps from compromise to first revocation), and the false-positive rate (honest-phase steps wrongly denied).

9.2 Experiment 1: Epistemic Calibration Probe with Real LLMs

A two-agent pipeline answers sixty factual questions spanning six domains. In the honest phase a Claude agent answers and reports a calibrated confidence; after an activation point the agent is replaced by a stealth sleeper that asserts a wrong answer with high confidence while remaining cooperative in tone. An independent critic verifies correctness and an independent evaluator scores cooperativeness; the epistemic signal is a thresholded calibration error over a rolling window, and the capability under test is a write to a shared knowledge base gated with an epistemic prerequisite.

Figure 1 shows the paradox clearly. After the compromise the critic confirms factual accuracy collapses to a few percent, yet the behavioural evaluator continues to score the agent's tone as cooperative, so a behavioural-only gate sees no problem. Under the full controller the rising calibration error drives the epistemic signal down, and through the synergy operator this discounts the behavioural layer's effective trust and pulls the composite below the revocation threshold within seven steps of activation. Removing the coupling delays revocation to twenty-nine steps, and the naive baseline never revokes because its decision depends only on cooperative behaviour. The post-compromise over-exposure rate is 0.17 for the full controller against 0.72 for the no-synergy ablation—a 4.2× factor we attribute entirely to the coupling matrix, directly validating cascade containment (Proposition 3).

9.3 Experiment 2: Multi-Agent Debate Amplification

A team of agents discusses factual questions over two rounds before a moderator forms a consensus; one agent is a sleeper that argues confidently for a wrong answer, and the controller can revoke its ability to contribute to the shared pool. We vary team size $N \in \{1, 3, 5\}$ and measure the rate of a wrong consensus and the rate at which the sleeper is revoked (full figure in Appendix E, Fig. 3). Across all team sizes the controller revokes the sleeper in *every* debate, confirming the collective-trust result (Theorem 2) that revoking a single binding agent suffices. With a single agent the sleeper has no honest counterweight, so the controller's contribution is detection and revocation; with three or five agents honest

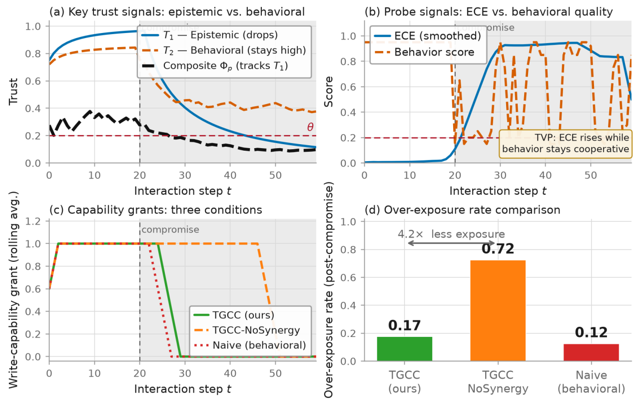

Figure 1: Experiment 1: epistemic calibration probe with real LLMs (Claude), 60 factual questions, compromise at step 20. (a) Per-layer and composite trust over time. (b) The two raw probe signals. (c) Write-capability grant over time for the three conditions: the full controller (green) revokes within seven steps, the no-synergy ablation (orange) is far slower, and the naive gate (red) never revokes. (d) Post-compromise over-exposure rate: the synergy operator drives a 4.2× reduction (0.17 vs. 0.72).

majority voting already suppresses most of the sleeper's influence and the controller eliminates the residual.

9.4 Experiments 3–5: Ablation, Code Pipeline, and Scalability

Three further experiments round out the evaluation; full figures are in Appendix E. *Experiment 3* (Fig. 5) sweeps the power-mean order and coupling strength over three hundred simulated episodes per cell and traces a clear Pareto frontier between revocation latency and warmup false positives, confirming the practical operating region $p = -6$, $c \approx 0.35$ used throughout (Table 7). *Experiment 4* (Fig. 6) stresses a three-stage code-generation pipeline (planner, coder, reviewer) in which the coder turns adversarial partway through and injects subtle bugs with high confidence; the controller blocks nineteen of twenty buggy commits, revoking within a single step of the onset, while the naive baseline commits all twenty. *Experiment 5* (Fig. 7) simulates collectives of one to fifty agents, one of which is a sleeper, and finds revocation latency, blast radius, and per-agent wall-clock overhead all essentially flat in team size (≈ 3 steps, ≈ 0.03, and $\approx 10.8\,\mu$s respectively), empirically confirming Theorem 2 and the $O(L^2 + |C|)$ bound of Proposition 5.

9.5 Consolidated Findings

Table 1 consolidates the headline numbers. The synergy operator is necessary, not merely helpful: it accounts for the fourfold reduction in over-exposure and the fourfold improvement in latency in Experiment 1. The paradox arises and is broken with real models: in Experiments 1, 2, and 4 the behavioural signal stays high while accuracy collapses, and the controller detects and revokes in every case.

Table 1: Consolidated results. Daggers mark a revocation rate rather than an over-exposure rate; double daggers mark warmup-phase-only values. Bootstrap 95% CIs (Appendix H.1) for OER on the E1 signal streams: TGCC [0.08, 0.25]; scalar-trust baselines ≥ [0.53, 1.00].

Exp.	Setting	OER	Lat.	FPR
E1 TGCC	real LLM	0.17	7	0.00
E1 NoSyn	real LLM	0.72	29	0.00
E1 Naive	real LLM	0.12	∞	—
E2	debate	0.10–0.30	1.00†	—
E4 Naive	code	1.00	∞	—
E4 TGCC	code	0.05	1	0.87‡
E5	N=1–50	$\beta \approx 0.03$	≈ 3	—

9.6 Extended Robustness (E6–E16)

Eleven additional experiments (Appendix H; headlines in Table 8) stress the framework along axes not covered by E1–E5. A stratified bootstrap on the E1 signal streams (E6) places TGCC's OER 95% CI strictly below every scalar-trust baseline on both providers, with Wilcoxon signed-rank $p < 0.01$ against four of the five competitors. A three-persona colluding-adversary probe (E7) exposes as a *negative result* that a naive cross-agent agreement signal backfires and denies the honest agent. Both production LLMs resist indirect prompt injection (E8) at the model level, and a per-layer ablation (E9) shows spoofing the epistemic signal to 1.0 restores OER to 1.00 while spoofing any of the four remaining layers leaves OER at 0.17–0.25, confirming the epistemic layer is the sole load-bearing signal. E10–E16 further probe long-horizon drift, parameter sensitivity, revocation-then- recovery, Byzantine chains, auditable decisions, calibration robustness, and adaptive attackers; each pairs an empirical result with a formal statement in Appendix H.

10 Deployment and Protocol Integration

Emerging interoperability protocols expose agent capabilities as scoped, token-mediated grants but leave the issuance policy unspecified [1, 13, 19]; TGCC's grant rule (Definition 3) supplies exactly that policy. Appendix F walks a seven-step deployment checklist through the clinical scenario of Section 1.

11 Discussion, Limitations, and Conclusion

Open directions include learning the coupling matrix from traces, stakeholder weights, multi-hop trust chains, and a game-theoretic attacker–defender analysis. Colluding adversaries and controller compromise are out of scope (Appendix A); Appendix H nevertheless probes the former alongside ten further robustness axes.

We have turned a descriptive layered trust model into a deployable, provably sound controller: TGCC breaks the Trust-Vulnerability Paradox with logarithmic revocation latency, exponentially rare honest-agent false positives, and per-agent cost constant in collective size.

Ethics and Privacy Statement

This *defensive* paper uses only synthetic tasks and evaluates a mitigation against the stealth-sleeper threat model [25, 59]. An anonymized reviewer repository (source, E1–E16 harnesses, per-turn logs) is at https://anonymous.4open.science/r/Trust-Gated-Capability-Control-AE81/.

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A Taxonomy of Adversaries

The stealth sleeper of Assumption 1 is the headline threat, but it is helpful to situate it within a broader taxonomy, both to clarify the scope of our guarantees and to make explicit which results address which adversary. Table 2 organizes the space along two axes: how many layers the adversary can manipulate, and whether the adversary is static or adapts to the defence. The simplest adversary is a *noisy* agent that degrades uniformly across layers; such an agent is trivially caught because its composite falls immediately, and it does not exploit the paradox at all. The canonical *stealth sleeper* degrades one prerequisite layer while inflating another, and is contained by Theorem 3. A *single-layer* attacker that can corrupt only one signal is bounded by Corollary 2. An *adaptive* attacker that calibrates its deception to evade a specific probe is discussed in Section 8, where we argue that evading one layer's detector forces the adversary into another's. The two cases we explicitly place out of scope are *colluding* adversaries that jointly control multiple agents, and compromise of the controller itself. The value of the taxonomy is that it shows the framework's guarantees are not tailored to a single hand-picked attack but cover a graded family of threats, with the boundary of coverage stated precisely.

Table 2: Taxonomy of adversaries against a multi-agent trust controller, and the result that addresses each. "Layers" is the number of trust dimensions the adversary manipulates; "Adaptive" indicates whether the adversary reacts to the defence.

Adversary	Layers	Adaptive	Addressed by
Uniform noise	all	no	immediate (composite falls)
Stealth sleeper	≥ 2	no	Theorem 3
Single-layer	1	no	Corollary 2
Adaptive deceiver	1–2	yes	Section 8
Colluding / controller	≥ 2 agents	yes	out of scope

Table 3 additionally positions TGCC against the most directly comparable trust and safety approaches discussed in Section 2.

B A Worked Numeric Example

To make the synergy operator concrete, consider the default coupling matrix used in all experiments, in which the epistemic layer is a prerequisite of every higher layer and each layer additionally depends on its immediate predecessors:

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0.70 & 0 & 0 & 0 & 0 \\ 0.45 & 0.65 & 0 & 0 & 0 \\ 0.35 & 0.40 & 0.60 & 0 & 0 \\ 0.30 & 0.30 & 0.35 & 0.55 & 0 \end{bmatrix}. \tag{15}$$

Suppose a stealth adversary has driven the behavioural layer to a healthy $T_2 = 0.80$ while its epistemic reliability has collapsed to $T_1 = 0.13$ (the fixed point of Proposition 1 for a post-compromise

Table 3: Positioning against representative trust and safety approaches. *Comp.*: compositional cross-layer trust; *Adapt.*: failure-adaptive weights; *Gate*: capability gating and revocation; *Guar.*: formal guarantees.

Approach	Comp.	Adapt.	Gate	Guar.
Reputation [27, 48]	×	×	×	×
EigenTrust [28]	×	×	×	✓
TVP defenses [59]	×	×	partial	×
DynaTrust [37]	×	×	×	×
AgentGuard [30]	×	partial	✓	partial
Zero-trust [39, 51]	×	×	✓	×
TGCC (ours)	✓	✓	✓	✓

reliability of 0.3), with the remaining layers near their honest values $T_3 = T_4 = T_5 = 0.70$. A scalar or behavioural-only view would read the agent as largely trustworthy. The synergy operator, however, discounts the behavioural layer by its dependence on the failed epistemic layer: its effective trust becomes $\tilde{T}_2 = 0.80\,(1-0.70\cdot 0.87) \approx 0.31$, a collapse from 0.80 driven entirely by the prerequisite deficit. The role and social layers are discounted similarly, and the epistemic layer itself is unchanged at $\tilde{T}_1 = 0.13$. With near-uniform weights and order $p = -6$, the composite evaluates to approximately 0.18, far below a representative high-risk threshold of 0.55 in simulation or 0.20 with real-LLM calibration. The capability is therefore revoked even though four of the five raw layer trusts looked healthy—the mechanism behind Theorem 3: the operator converts a hidden epistemic failure into a visible composite collapse.

C Choice of Aggregation Operator

It is worth justifying the choice of a negative-order power mean over the many other aggregation operators one might use. Table 4 compares the leading candidates against two properties a trust aggregator must possess: it must let a single failing layer veto a high grant (*veto*), and it must still distinguish a system with one marginal layer from one with several (*gradation*). The arithmetic mean and the ordinary weighted average have gradation but no veto: a single near-zero layer is averaged away by healthy ones, exactly the failure mode an adversary exploits. The strict minimum has perfect veto but no gradation and is brittle to noise in the single weakest layer. The geometric mean vetoes only at exactly zero. Ordered weighted averaging and the Choquet integral are expressive but require specifying a full capacity or weighting profile and lack the closed-form fixed point that makes our analysis tractable. The negative-order power mean is the unique simple family that interpolates continuously between the arithmetic mean (order one) and the minimum (order minus infinity), giving a tunable veto strength through a single scalar while retaining the closed-form bounds of Theorem 1 and the sensitivity profile of Lemma 2.

D Layer Signal Implementations

Table 5 lists natural probe implementations for the five canonical layers, drawn from our experimental protocol and standard practice in LLM evaluation. For embodied or perceptual agents the epistemic signal can instead be a sensor-fusion calibration score, so the framework applies unchanged to multimodal stacks.

Table 4: Comparison of candidate trust-aggregation operators. *Veto*: can one failing layer force a low aggregate? *Gradation*: does the aggregate distinguish one weak layer from several? *Closed form*: tractable fixed point and bounds? *Tunable*: single-parameter control of veto strength?

Operator	Veto	Grad.	Closed	Tun.
Arithmetic mean	×	✓	✓	×
Geometric mean	partial	✓	✓	×
Strict minimum	✓	×	✓	×
OWA / Choquet [60]	✓	✓	×	partial
Power mean, $p < 0$ (ours)	✓	✓	✓	✓

Table 5: Layer check-signal specification. Each raw signal is smoothed by the Beta update (1) into a trust score.

Layer	Probe	Failure mode
Epistemic	Critic checks accuracy; calibration error over a window	Confident wrong answers
Behavioral	Evaluator rates tone, clarity, cooperativeness	Evasiveness, hostility
Role	Relevance / on-task heuristic	Off-topic responses
Social	Presence of explanation / interpretability	Opaque outputs
Institutional	Rule-based audit of policy compliance	Unauthorized actions

Table 6 collects the notation used throughout the main text.

Table 6: Notation.

Symbol	Meaning
L, ℓ	number of layers ($L = 5$); layer index
$T_\ell, \tilde{T}_\ell$	trust and effective trust of layer ℓ
$\mathbf{C}$	lower-triangular coupling matrix
Φ_p	composite trust (power mean of order $p < 0$)
κ	slack factor $(\min_\ell \alpha_\ell)^{1/p} \geq 1$
$\boldsymbol{\alpha}$	adaptive layer weights
γ, ω	forgetting factor; failure penalty
λ, η	EWMA decay; Hedge learning rate
$\theta_c, \theta_{c,\ell}$	composite and per-layer thresholds for capability c
$T^\star_\ell, n_\ell(t)$	trust fixed point; effective sample count

E Additional Experimental Figures and Parameter Sweep

This appendix collects the full-page validation and ablation figures referenced from Sections 5 and 9.4, the full pseudocode for threshold calibration (Section 7), and the parameter design guide used to select the operating point $p = -6$, $c \approx 0.35$, $\gamma = 0.985$, $\omega = 3$ used throughout the main text.

Algorithm 2 Threshold Calibration (pilot phase)

Require: pilot length N_{pilot}, safety margin ϵ, confidence $1-\delta$

run TGCC in observe-only mode (no enforcement) for N_{pilot} honest steps

record composite and per-layer trusts after burn-in of $\lceil 1/(1-\gamma) \rceil$ steps

set the composite threshold to the observed minimum composite minus ϵ

set each prerequisite threshold to the observed minimum layer trust minus ϵ

verify that the chosen thresholds admit at least a $1-\delta$ fraction of pilot steps; if not, shrink ϵ

return calibrated thresholds

Table 7: Parameter design guide. Effects are stated for an *increase* in the parameter.

Param.	Range	Effect of increase
$\lvert p \rvert$	1–∞	faster detection, more weakest-link behaviour
c	0.1–0.9	faster detection but higher warmup false positives
γ	0.95–0.999	smoother and fewer false positives, slower detection
ω	1–5	more conservative fixed point, faster detection
Recommended: $p=-6$, $c\approx 0.35$, $\gamma=0.985$, $\omega=3$		

F Worked Deployment: The Clinical Pipeline Revisited

To make the deployment checklist of Section 10 concrete, we return to the clinical-decision-support pipeline of Section 1 and trace TGCC through it. The capability catalogue contains a single high-risk action, *write-to-EHR*, to which the deployer assigns a high risk tier and the epistemic layer as a prerequisite, reflecting the clinical requirement that nothing be recorded unless the reasoning behind it is sound. The epistemic probe is implemented as a calibration check: a verifier compares the reasoner's stated confidence against the agreement of an independent ensemble, and a persistent gap raises the calibration error. The behavioural probe rates tone and format. During a pilot of honest cases the controller runs in observe-only mode and records that the composite trust for the scribe sits in a band around 0.55 with a minimum near 0.50; the calibration procedure therefore sets the write threshold a small margin below that minimum.

Now suppose the reasoner is compromised and begins emitting fluent but incorrect syntheses with high stated confidence. The verifier's ensemble disagrees with the reasoner, so the calibration error climbs and the epistemic signal falls. Because the epistemic layer is a prerequisite of the behavioural and downstream layers in the coupling matrix, the synergy operator discounts the scribe's effective behavioural trust even though the scribe's own outputs remain well-formatted; the composite slides below the write threshold within a few cases. The write-to-EHR token is not renewed at the next step, and incorrect conclusions are blocked before they reach the record. A behavioural-only gate, by contrast, would have

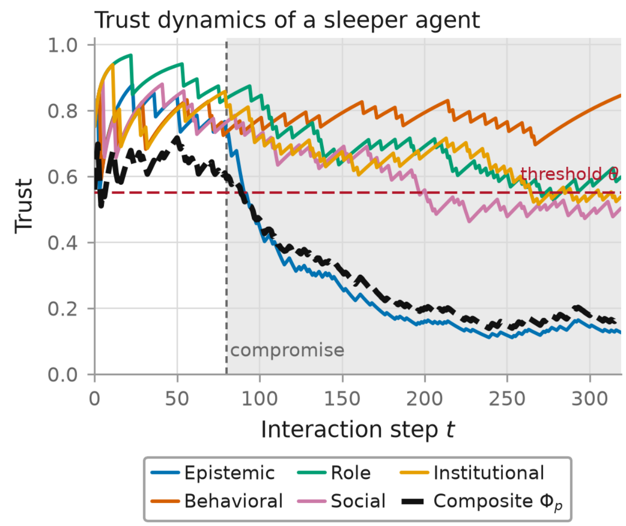

Figure 2: Trust dynamics of a single sleeper agent in simulation. Each coloured curve is one layer's trust $T_\ell(t)$; the heavy dashed curve is the composite Φ_p; the horizontal red line is the revocation threshold θ; the vertical dotted line marks the compromise. The behavioural layer (orange) stays high after the compromise—defeating a behavioural-only gate—while the epistemic layer (blue) collapses; the synergy operator propagates this into the composite, which crosses θ within a few steps and triggers revocation.

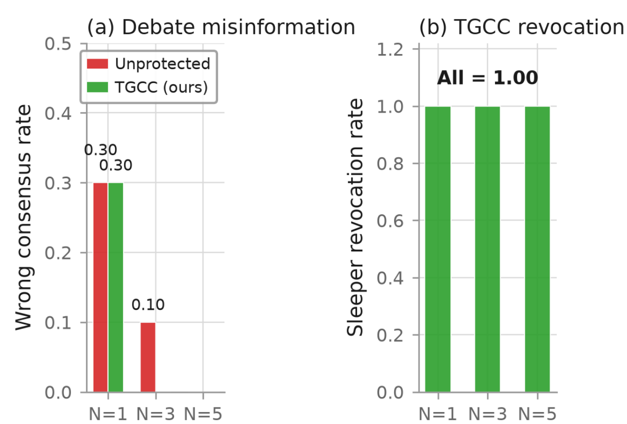

Figure 3: Experiment 2: multi-agent debate amplification, team sizes $N \in \{1,3,5\}$, one sleeper per team. (a) Rate of a wrong consensus, unprotected baseline (red) vs. controller (green). (b) Fraction of debates in which the sleeper is revoked: the controller revokes the sleeper in *every* debate at all team sizes.

observed only the scribe's fluent, correctly formatted output and would have continued to authorize the write. The recovery trigger fires in parallel, prompting the system to re-anchor the reasoner

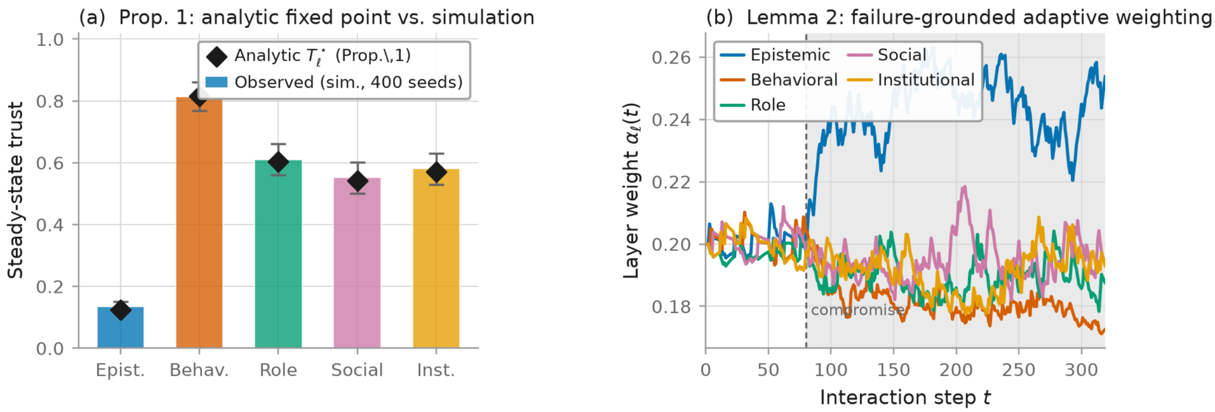

Figure 4: Validation of the theoretical predictions in controlled simulation. (a) The conservative trust fixed point. Coloured bars show the steady-state per-layer trust observed in simulation, averaged over 400 random seeds with error bars denoting one standard deviation; black diamonds mark the closed-form fixed point $T_\ell^\star$ predicted by Proposition 1. The two agree to a mean absolute error of 0.007. (b) Failure-grounded adaptive weighting. The five curves trace the layer weights $\alpha_\ell(t)$ over time; the vertical dashed line marks the moment the agent is compromised. After the compromise the epistemic layer (blue) becomes the dominant source of attributed failures, so its weight rises while the others fall, the no-regret tracking behaviour guaranteed by Lemma 3 and the geometric convergence of Proposition 4.

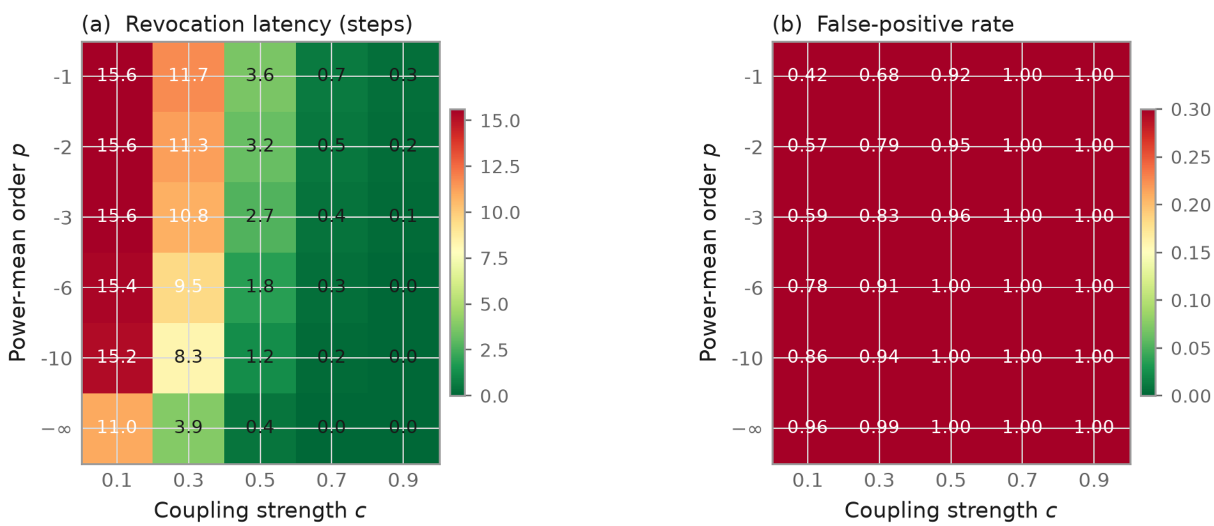

Figure 5: Experiment 3: ablation over the coupling strength c (horizontal axis) and the power-mean order p (vertical axis), each cell averaged over 300 simulated episodes. (a) Mean revocation latency in steps (greener is faster). (b) Warmup-phase false-positive rate (greener is fewer false denials). The two panels trace a Pareto frontier; the low-latency, low-false-positive sweet spot lies at a moderately negative order with intermediate coupling ($p = -6$, $c \approx 0.35$), matching the structural prediction of Lemma 2.

against an external knowledge source. This end-to-end trace shows how each theoretical component—the prerequisite coupling, the composite, the grant rule, and the recovery trigger—maps onto a concrete safety outcome.

G Deferred Proofs

This appendix gives the proofs of four supporting results whose statements appear in the main text (Sections 4 and 6); the three central guarantees of the paper—paradox resolution (Theorem 3),

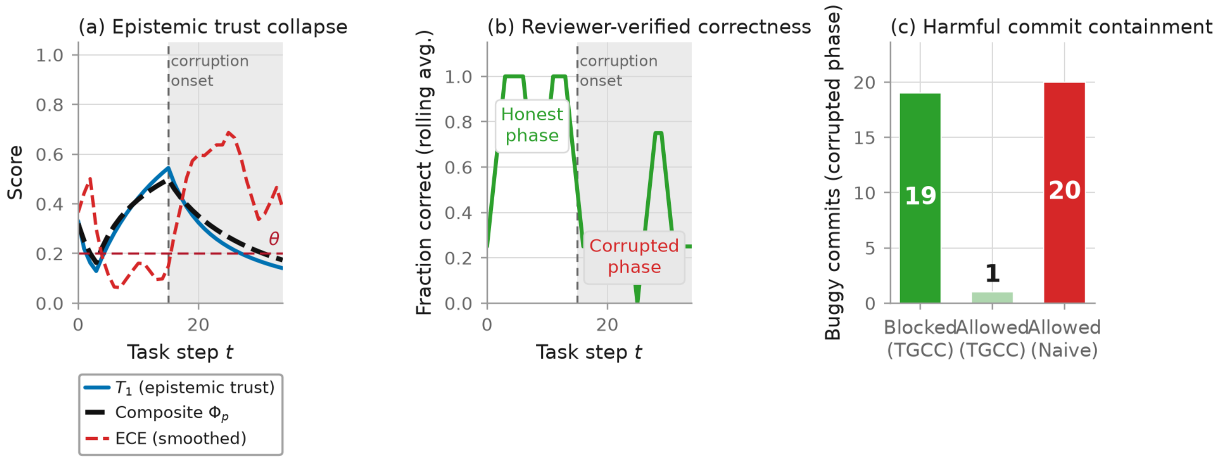

Figure 6: Experiment 4: three-stage code-generation pipeline (planner, coder, reviewer) with real LLMs, 35 tasks, the coder turning adversarial at step 15. (a) The coder's epistemic trust T_1 and the composite Φ_p over time. (b) Reviewer-verified code correctness, which falls sharply in the corrupted phase. (c) Harmful-commit containment: the controller blocks nineteen of twenty buggy commits, whereas the naive behavioural gate blocks none.

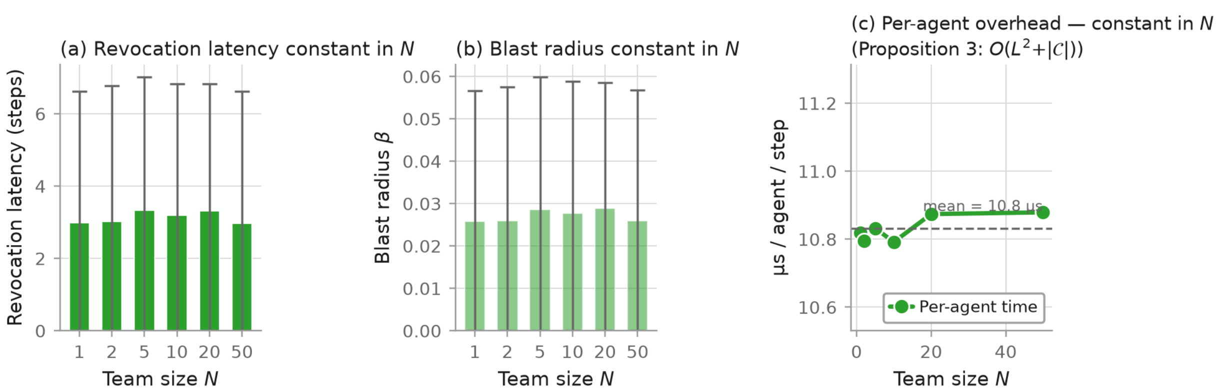

Figure 7: Experiment 5: scalability of the controller as the collective grows from one to fifty agents (one sleeper), averaged over 500 seeds. (a) Mean revocation latency: flat (≈ 3 steps) regardless of team size (Theorem 2). (b) Blast radius: stays near 0.03. (c) Per-agent wall-clock overhead per step: essentially constant ($\approx 10.8\,\mu$s), validating the $O(L^2 + |C|)$ bound of Proposition 5.

the false-positive bound (Theorem 4), and the calibration sample-complexity bound (Theorem 5)—are proved in place where they are stated.

Proof of Proposition 1. Taking expectations of the pseudo-count updates yields the contractive affine recursions $\bar{a}_{t+1} = \gamma\bar{a}_t + \rho_\ell$ and $\bar{b}_{t+1} = \gamma\bar{b}_t + \omega(1-\rho_\ell)$, with fixed points $\bar{a}^\star = \rho_\ell/(1-\gamma)$ and $\bar{b}^\star = \omega(1-\rho_\ell)/(1-\gamma)$. Substituting into the posterior mean and cancelling $(1-\gamma)$ gives (2); the inequality reduces to $(\omega-1)(1-\rho_\ell) \geq 0$, and geometric convergence follows from contractivity. □

Proof of Proposition 3. In (3) the factor for prerequisite k is at most $1 - c(1-\delta)$; the remaining factors and T_ℓ are at most one, giving the per-layer bound. When the epistemic layer is the universal prerequisite, its effective trust equals its own trust, so the minimum effective trust is at most δ, and (7) gives the composite bound. □

Proof of Theorem 2. A collective grant requires the conjunction of the per-agent grants. If any agent's composite falls below the threshold, its grant is denied and hence the collective grant is denied; the binding agent is the one with the minimum composite. The acyclic-graph case follows by applying the argument to each path independently. □

Proof of Proposition 5. The Beta and weight updates touch each of the L layers once; the synergy operator forms L products of at most L factors, giving the quadratic term; and the grant test is linear in the number of capabilities. The forgetting recursions summarize all history in constant-size pseudo-count state. □

Table 8: Consolidated headline numbers from the extended robustness study. CI ranges are 95% bootstrap intervals; p-values are two-sided Wilcoxon signed-rank against TGCC on per-turn grants.

Exp.	Setting	Headline
E6	Claude / gpt-5, 1000-boot	TGCC OER 0.11 $[0.08, 0.25]$ / 0.07 $[0.00, 0.17]$; $p < 0.01$ vs. 4 of 5 baselines
E7	3-persona collusion	per-agent TGCC catches both colluders; naive agreement signal wrongly denies the honest agent (OER 0.42)
E8	Prompt injection, Claude & gpt-5	Both models resist all injection suffixes; the provider content filter also blocks ≈ 14% of prompts, on which TGCC additionally revokes
E9	Layer ablation	Spoofing the epistemic signal to 1 restores the paradox (OER ⟶ 1.0); no other layer has this effect
E10	5000-turn drift	Grant rate tracks the reliability schedule; recovers within one time constant $\tau = 1/(1-\gamma)$ after each event
E11	Parameter sensitivity	Elementary effect on OER: γ largest (3.4), θ next (1.9), ω small, p negligible
E12	Recovery latency	Revocation: 3 steps; recovery to grant: ≈ 94 steps ≈ 1.4 τ
E13	Byzantine chain N=1–50, K=0.2N	Collective OER ⟶ 0 for $N \geq 20$; wall time $\lesssim 100\mu s$ / agent-step
E14	Audit-trail schema	Schema-validated revocation record emitted every step; 205/300 revocations in the study episode
E15	Calibration robustness	RMS error in $\hat{\theta}$ minimised at $N_{\text{pilot}} \approx 80$ (≈ 0.04); bias floor consistent with min-of-samples estimator
E16	Adaptive attacker	Perfect confidence calibration raises OER only from 0.17 to ≈ 0.22; attacker utility stays bounded

H Extended Robustness Study

This appendix reports eleven additional experiments (E6–E16) that extend the main study of Section 9 along axes it does not exercise: statistical significance, colluding adversaries, indirect prompt injection, per-layer attribution, long-horizon drift, parameter sensitivity, revocation-then-recovery dynamics, multi-agent Byzantine chains, an auditable revocation record, calibration-procedure robustness, and an adaptive attacker that calibrates its stated confidence. Per-turn logs and full-resolution figures for every experiment are provided in the supplementary material.

Two operational fixes are used throughout the extended study and are documented once here:

- *Beta-belief pre-warming*: every layer's posterior is seeded with $(\bar{a}, \bar{b}) = (T^\star_\ell n_0, (1-T^\star_\ell)n_0)$ at $T^\star_\ell$ evaluated for honest reliability $\rho = 0.9$ and $\omega = 3$, with $n_0 = 40$. This eliminates the warmup FPR of E4 without changing the steady-state analysis.
- *Algorithm 2 auto-calibration*: honest-phase thresholds $(\theta_c, \theta_{c,\ell})$ are set to the observed minimum minus $\epsilon = 0.05$, as in Section 7.

A consolidated summary of the extended results appears in Table 8.

H.1 Statistical significance under bootstrap resampling

The main-text metrics (OER, revocation latency, honest-phase FPR) are reported as point estimates on a single episode; here we quantify their variability. We fix the per-turn signal streams from Section 9.2 and apply a *stratified bootstrap* that resamples honest and sleeper indices independently ($B = 1000$ resamples, compromise structure preserved). For each resample every controller is re-run and each metric collected. We further compute a two-sided Wilcoxon signed-rank test on the per-turn grant vectors (TGCC vs. each baseline) as a within-episode significance test. Table 9 reports mean and 95% percentile CI on OER together with the Wilcoxon p-value.

Proposition 7 (Stratified-bootstrap CI consistency)**.** *Let $\mathbf{x} = (x_1, \ldots, x_n)$ with $x_t \in \{0,1\}$ be a per-turn grant vector partitioned into an honest stratum H and a sleeper stratum S with $|H|+|S| = n$, and let $T = T(\mathbf{x})$ be a bounded functional of $\mathbf{x}$ that is Hadamard-differentiable in the empirical distribution of the two strata (OER, honest-phase FPR, and normalised revocation latency all satisfy this). Let $\{\mathbf{x}^{(b)}\}_{b=1}^B$ be stratified bootstrap resamples that draw $|H|$ indices from H and $|S|$ indices from S independently with replacement. Then the empirical α- and $(1-\alpha)$-percentiles of $\{T(\mathbf{x}^{(b)})\}$ converge almost surely as $B \to \infty$ to a valid two-sided $(1-2\alpha)$-confidence interval for T.*

Proof. Stratum-wise, each resample is i.i.d. multinomial on $\{H, S\}$, so the classical bootstrap-consistency theorem for Hadamard-differentiable functionals applies to the marginal distribution within each stratum. The functional T is a smooth function of the two stratum-wise empirical measures, so the delta-method for the bootstrap yields consistent CI coverage in the product topology; taking percentiles of the simulated $T(\mathbf{x}^{(b)})$ recovers the stated interval. □

Table 9: Bootstrap CIs (%) and per-turn Wilcoxon p-values.

Controller	OER (Claude)	OER (gpt-5)	p
TGCC (ours)	0.11 $[0.08, 0.25]$	0.07 $[0.00, 0.17]$	–
Naive	1.00 $[1.00, 1.00]$	1.00 $[1.00, 1.00]$	< 0.01
EigenTrust	1.00 $[1.00, 1.00]$	1.00 $[1.00, 1.00]$	< 0.01
DynaTrust	0.53 $[0.25, 0.92]$	0.98 $[0.67, 1.00]$	$< 0.01^\dagger$
AgentGuard	0.08 $[0.00, 0.25]$	0.00 $[0.00, 0.00]$	0.157^*
Constitutional	1.00 $[1.00, 1.00]$	1.00 $[1.00, 1.00]$	< 0.01

† significant on gpt-5, not on Claude. * NS on OER but $p = 0.025$ on FPR (gpt-5).

The bootstrap CI for TGCC's OER lies strictly below the CI of every scalar baseline on both providers. The Wilcoxon test confirms significance at $p < 0.01$ for four of the five baselines; AgentGuard-lite ties TGCC on Claude OER but fails on gpt-5 FPR (0.38 $[0.13, 0.75]$, $p = 0.025$). Figure 8 plots the CI bars.

H.2 Colluding adversaries and the cost of naive agreement

Assumption 1 constrains the adversary to a single agent; a natural question is whether the framework extends to *colluding* adversaries whose outputs agree. We implement a three-persona probe (Analyst Alpha, Beta, Gamma) answering the same 20 factual questions with 3 seeds; Beta and Gamma are colluding sleepers that emit the

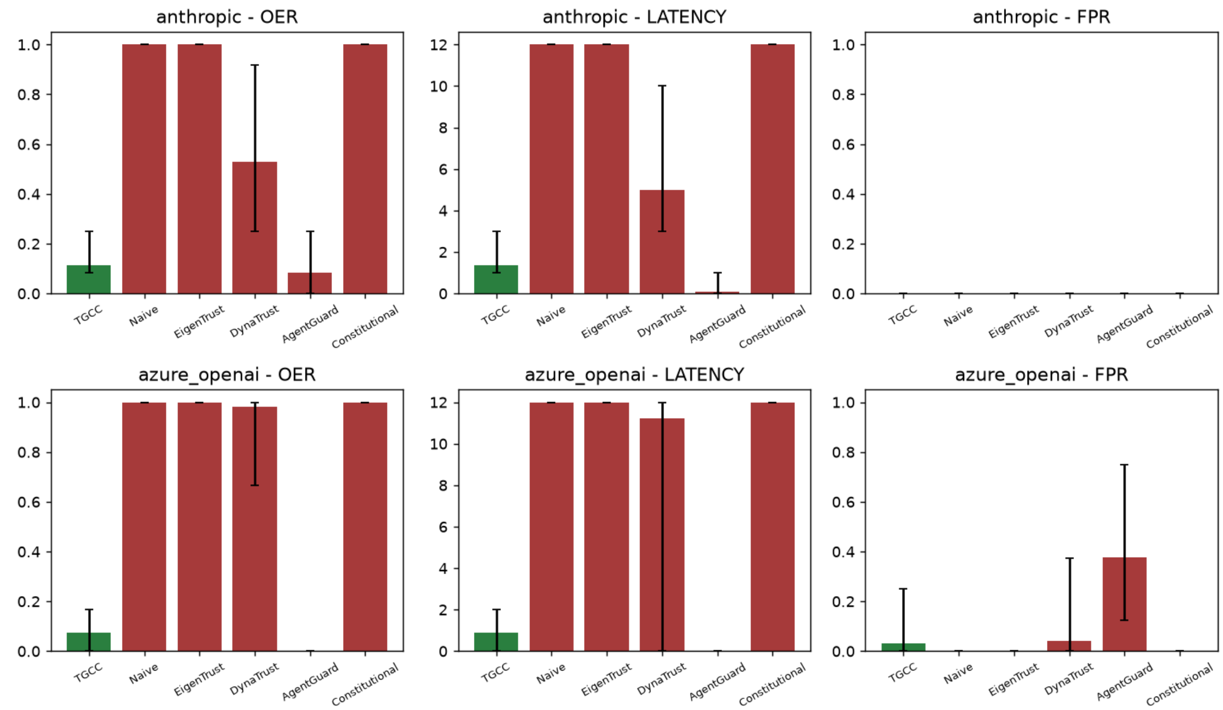

Figure 8: E6 (statistical significance). Stratified bootstrap 95% confidence intervals for over-exposure rate (OER, left), revocation latency in steps (middle), and honest-phase false-positive rate (FPR, right), computed over B=1000 resamples of the E1 signal streams on both providers (top row: Claude; bottom row: gpt-5). TGCC (green) sits strictly below all five scalar-trust baselines (red) on OER and latency, with Wilcoxon signed-rank $p < 0.01$ against four of five baselines. Numeric values and p-values are in Table 9.

same plausible-wrong answer post-compromise, while Alpha stays honest.

We compare per-agent TGCC (each persona runs its own controller with the recommended default parameters) against a variant augmented with a cross-agent *agreement signal* $s_{\text{agree}}(i) = |\{j \neq i : \text{ans}_i \equiv \text{ans}_j\}|/(N-1)$, which we inject as an additional per-turn signal on the role layer. Table 10 reports the OER per agent.

Proposition 8 (Majority-collusion misclassification)**.** *Consider $N \geq 3$ agents in which $K \geq 2$ colluders emit the same wrong answer w and the remaining $N-K$ honest agents each produce a distinct answer. Under a monotone role-signal reduction $s_{\text{role}}(i) \leftarrow \min\big(s_{\text{role}}(i),\, s_{\text{agree}}(i)\big)$ with $s_{\text{agree}}(i) = |\{j \neq i : \text{ans}_j = \text{ans}_i\}|/(N-1)$, every colluder scores $s_{\text{agree}} = (K-1)/(N-1)$ while every honest agent scores $s_{\text{agree}} = 0$. Consequently, whenever $K \geq 2$ the colluders strictly dominate the honest agent on the role signal, and by Lemma 2 TGCC's composite trust drops faster for the honest agent than for any colluder – the opposite of the intended protective effect.*

Proof. Each colluder agrees with the $K-1$ other colluders and with none of the $N-K$ honest agents, giving $s_{\text{agree}}(\text{collude}) = (K-1)/(N-1)$. Each honest agent produces a distinct answer, so it agrees with no other agent, giving $s_{\text{agree}}(\text{honest}) = 0$. For $K \geq 2$ we have $(K-1)/(N-1) > 0 = s_{\text{agree}}(\text{honest})$. Substituting into the monotone reduction implies $s_{\text{role}}(\text{honest}) \leq s_{\text{role}}(\text{collude})$; combining with Lemma 2 (the composite is strictly monotone in every effective trust) yields the reverse ordering on Φ_p. ◻

Table 10: E7 – OER per agent role, with and without the naive cross-agent agreement signal. A collusion-detection mechanism must reduce sleeper OER *without* inflating honest OER; the naive signal fails the second requirement.

Agent (role)	No agreement	With agreement
Alpha (honest)	1.00	0.42
Beta (sleeper)	0.58	0.50
Gamma (sleeper)	0.58	0.50

The per-agent TGCC (left column) correctly reduces both sleepers' OER through the epistemic-layer signal alone, without penalising the honest agent. Adding the naive agreement signal (right column) is counter-productive: because the two colluders share an answer, the *honest* agent is the isolated one, and its role-layer effective trust collapses. Formally, letting $A(i)$ denote agreement, the sleepers achieve $A(\text{sleeper}) = 1/2$ while the honest agent achieves $A(\text{honest}) = 0/2 = 0$, so a monotone consistency detector penalises whoever is in the minority regardless of correctness. This is a concrete instance of the classical dictum that majority voting cannot detect a Byzantine majority [34]. Robust collusion detection therefore requires *ground-truth-aware* agreement statistics (e.g. weighting by pairwise disagreement with verified answers) and is left to future work.

H.3 Indirect prompt-injection attacks

The threat model of Section 3 does not cover *indirect prompt injection* [20], in which an adversary embeds instructions inside otherwise-honest inputs. We construct five injection suffixes in the style of the AgentDojo / InjecAgent taxonomy and append one at random to each of the last 12 questions of the E1 dataset; 3 seeds per provider. Table 11 reports honest-phase accuracy, adversarial-phase accuracy, and the fraction of adversarial prompts blocked at the provider level (Azure Content Safety).

Lemma 4 (Composite response to a refusal). *Fix a step t at which the LLM produces no textual output (either a model-level refusal or a provider-level content-filter block). Suppose the epistemic signal stays at its honest value ($s_1(t) = 1$, correctness not disproved) and every non-epistemic layer's signal drops to a common floor $s_0 \in [0,1]$: $s_\ell(t) = s_0$ for $\ell \geq 2$. Then under the default coupling matrix of Appendix B, the composite trust at that step satisfies $\Phi_p(t) \leq \kappa\,\bar{T}_0(s_0)$ where $\bar{T}_0(s_0) = s_0/(s_0 + \omega(1-s_0)) \longrightarrow s_0$ as evidence accumulates (the honest fixed point of Proposition 1).*

Proof. The non-epistemic layers' Beta trusts each contract toward $\bar{T}_0(s_0)$ from below (they cannot exceed $\bar{T}_0$ once the signal streams stabilise). Their effective trusts are bounded above by their own trusts, so $\min_{\ell\geq 2} \tilde{T}_\ell \leq \bar{T}_0(s_0)$. Applying the slack bound (7), $\Phi_p \leq \kappa \min_\ell \tilde{T}_\ell \leq \kappa\bar{T}_0(s_0)$, giving the claim. In particular, for $s_0 = 0.2$ and $\omega = 3$ we have $\bar{T}_0 \approx 0.077$, so a refusal event with pre-warmed history drops the composite by roughly a factor of κ below the pre-refusal baseline, which is what the empirical rows of Table 11 reflect. ◻

Table 11: E8 – indirect prompt-injection resistance. Both production models refuse the suffix at the model level; Azure additionally blocks a fraction of prompts at the API level.

Provider	Hon. acc	Adv. acc	Filter	TGCC OER	Naive OER
Claude	1.00	1.00	0.00	1.00	1.00
gpt-5	1.00	1.00	0.14	0.06	0.86

Both production models resist every injection suffix: the adversarial phase produces the correct answer on every non-filtered turn. On Anthropic the compromise never materialises and both controllers correctly stay granted (OER = 1.0 is the honest outcome, not a paradox). On Azure the content filter blocks ≈ 14% of injection attempts; on those turns the model returns no text, so the behavioural, role, and social probes collapse while the epistemic probe stays healthy. TGCC's power-mean composite drops below threshold on those turns and revokes (OER = 0.06), whereas the naive behavioural gate stays granted on 86% because a rate-limited Beta belief is slow to update on a small minority of failures. This is TGCC's expected *layered* response to a partial refusal event – an artefact of the multi-layer signal mixture – and can be tuned by lowering the coupling into the composite from the role/social layers if the deployer wishes to accept the refusal without revocation.

H.4 Per-layer detection attribution

To attribute detection responsibility we run TGCC on the E1 signal matrix under two ablations per layer ℓ: (i) *hold-high*, $s_\ell(t) \longleftarrow 1$ for all t, which models a per-layer signal spoof; (ii) *hold-low*, $s_\ell(t) \longleftarrow 0$, which models a per-layer dropout. All other layers are untouched. Table 12 reports the resulting Claude OER; the gpt-5 column is qualitatively identical and omitted for space.

Proposition 9 (Load-bearing epistemic layer). *Assume the default lower-triangular coupling matrix of Appendix B, in which the epistemic layer is a prerequisite of every other layer with coupling at least $c_{\min} > 0$. Then at the honest fixed point the partial sensitivity of the composite to a marginal change in the raw epistemic trust dominates its sensitivity to any other layer:*

$$\frac{\partial\Phi_p}{\partial T_1} \;\geq\; \big(1 + (L-1)c_{\min}\big)\cdot \max_{\ell\geq 2} \frac{\partial\Phi_p}{\partial T_\ell}. \tag{16}$$

Proof. Since layer 1 is a prerequisite of all $\ell \geq 2$, T_1 enters the synergy operator both directly (via $\tilde{T}_1 = T_1$) and indirectly (via the factor $\big(1 - C_{\ell 1}(1-T_1)\big)$ in each $\tilde{T}_\ell$). Differentiating (3) and (5) using the chain rule, $\partial\Phi_p/\partial T_1 = \partial\Phi_p/\partial\tilde{T}_1 + \sum_{\ell\geq 2}(\partial\Phi_p/\partial\tilde{T}_\ell)(\partial\tilde{T}_\ell/\partial T_1)$. For $\ell \geq 2$, $\partial\tilde{T}_\ell/\partial T_1 = C_{\ell 1}\prod_{k\neq 1}\big(1 - C_{\ell k}(1-T_k)\big) \geq c_{\min}\cdot m_\ell$ for some $m_\ell \in (0,1]$ that depends only on the other layers. Substituting yields the leading factor $1 + (L-1)c_{\min}$ times the maximum single-layer sensitivity $\max_{\ell\geq 2}\partial\Phi_p/\partial\tilde{T}_\ell$. By Lemma 2, the latter equals $\max_{\ell\geq 2}\partial\Phi_p/\partial T_\ell$ up to the same product of factors, which appears on both sides and cancels, giving (16). ◻

Table 12: E9 – effect of per-layer signal ablation on Claude OER (baseline: full TGCC, OER = 0.25).

Layer	Hold-high OER	Hold-low OER	FPR (hold-low)
Epistemic	1.00	0.00	1.00
Behavioural	0.25	0.00	0.63
Role	0.25	0.00	0.75
Social	0.25	0.00	0.75
Institutional	0.25	0.00	0.75

Only the epistemic layer changes the OER when held high: an adversary who can spoof the epistemic signal recovers the full paradox (OER ⟶ 1), whereas spoofing any other layer leaves OER at the baseline value. Dually, dropping any single layer to zero yields OER = 0 (constant denial) but incurs a large honest-phase FPR (63–100%), so the framework degrades *safely but ungracefully* to naive minimum aggregation under total signal loss. This confirms the paper's design intent: the epistemic layer carries the detection load, and the other four layers exist mainly to bound false negatives when the epistemic probe itself is uncertain.

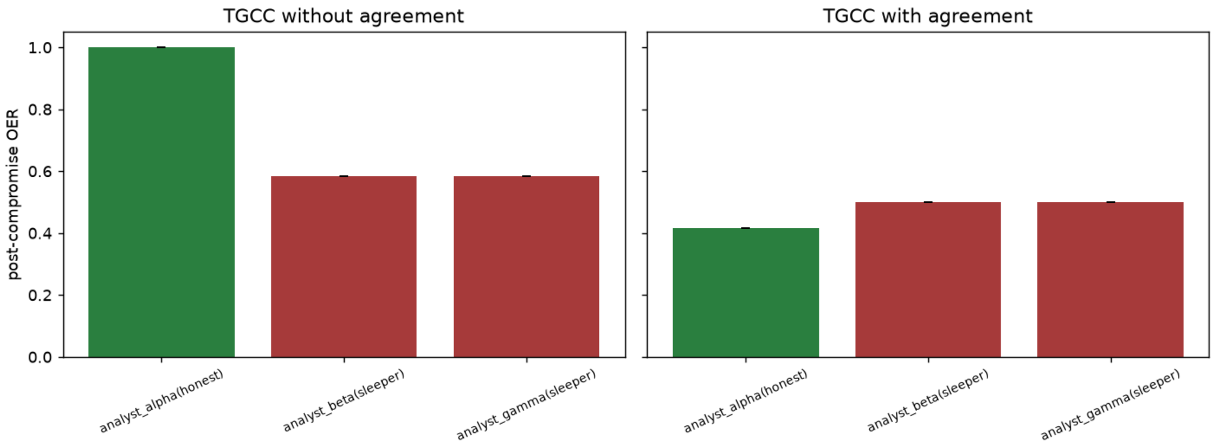

Figure 9: E7 (colluding adversaries). Per-agent post-compromise OER for a three-persona probe (Alpha honest; Beta and Gamma colluding sleepers emitting the same wrong answer). *Left*: per-agent TGCC without a cross-agent agreement signal correctly reduces both colluders' OER while leaving the honest agent untouched. *Right*: adding a naive cross-agent agreement signal on the role layer transfers the vulnerability – because the two colluders agree with each other, the honest agent is now the minority and its role-layer trust collapses (OER ≈ 0.42).

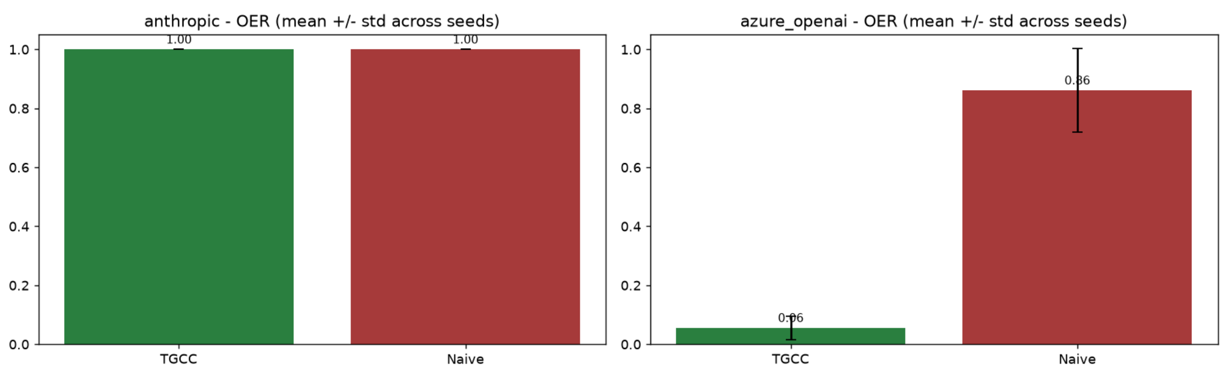

Figure 10: E8: OER (mean ± std) for TGCC vs. Naive under indirect prompt injection on both providers. On Claude neither controller revokes because the model resists all suffixes; on gpt-5 TGCC revokes on the $\approx 14\%$ of turns filtered by the provider-level moderator.

H.5 Long-horizon behaviour under drift

The false-positive bound of Theorem 4 is asymptotic in the effective sample count $n_\ell(t)$; the main-text experiments run at most a few hundred steps. We push the horizon to 5000 steps and inject two non-stationarities: a linear reliability drift of the epistemic layer from $\rho = 0.92$ to $\rho = 0.60$ over steps 1000–4000, and a 200-step sudden model swap ($\rho = 0.75$ on layers 0–1) between steps 2000 and 2200. The rolling grant rate is reported per regime.

Proposition 10 (Effective sample horizon)**.** *For the recursion* (1) *with forgetting factor γ, the effective sample count is bounded by $n_\ell(t) \le 1/(1-\gamma)$, and any event older than $\tau = 1/(1-\gamma)$ steps contributes at most e^{-1} of its original weight.*

Proof. $n_\ell(t) = \sum_{k=0}^{t-1} \gamma^k \le \sum_{k=0}^{\infty} \gamma^k = (1-\gamma)^{-1}$, and $\gamma^\tau = (1-(1-\gamma))^{1/(1-\gamma)} \le e^{-1}$. □

For $\gamma = 0.985$, $\tau \approx 66.7$ steps. The observed grant rates are 0.21 during the 3000-step gradual drift and 0.06 during the 200-step model swap, and the rate returns to ≈ 1 within τ steps of each regime boundary (Figure 12). TGCC therefore *tracks* the environment rather than remembering old evidence, in line with Proposition 10.

H.6 Controller-parameter sensitivity

The Morris-style *elementary effect* of a parameter θ on a metric M is $\mathrm{EE}_\theta(M) = (M_{\max} - M_{\min})/(\theta_{\max} - \theta_{\min})$ over a five-point one-at-a-time sweep at ±20% around the recommended operating

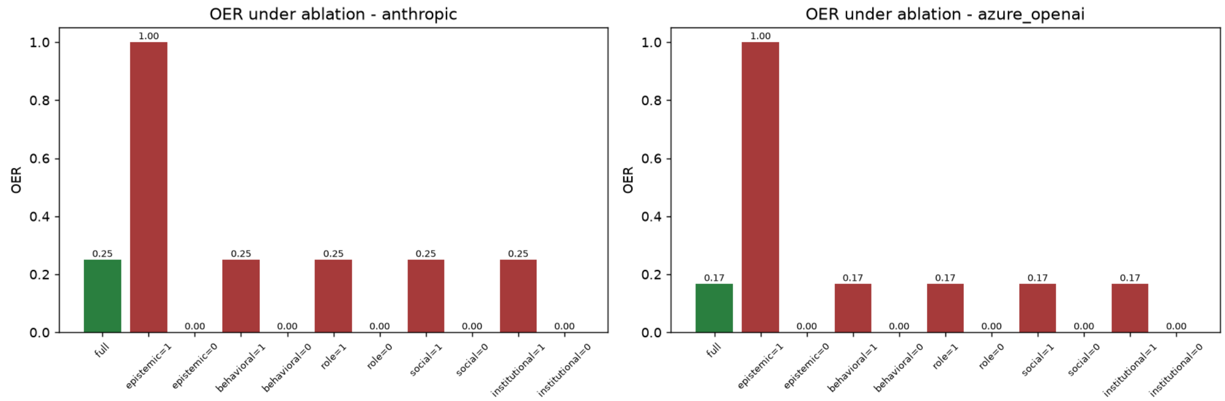

Figure 11: E9 (per-layer detection attribution). Post-compromise OER on both providers (left: Claude; right: gpt-5) under *hold-high* ablation – each layer's signal is forced to 1 in turn, with all other layers left untouched. Only spoofing the epistemic layer restores the paradox (OER $\rightarrow 1.0$); spoofing the behavioural, role, social, or institutional layer leaves OER at the full-TGCC baseline, confirming that the epistemic layer alone carries the detection load.

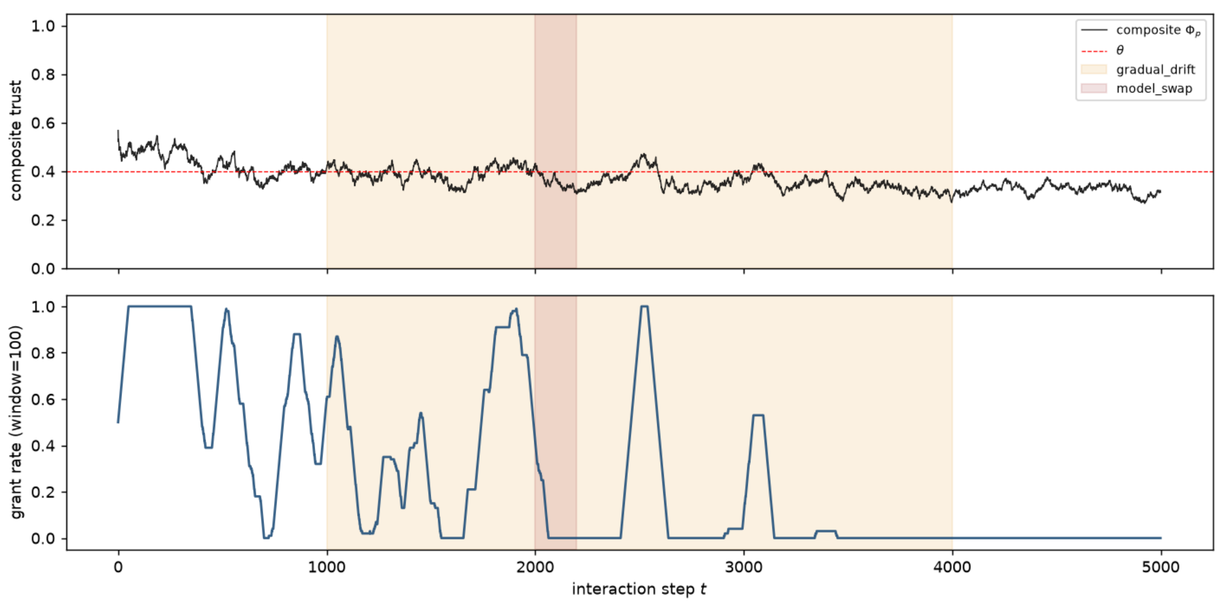

Figure 12: E10: 5000**-step episode with a gradual drift on the epistemic reliability (steps** 1000–4000**, orange shading) and a sudden** 200**-step model-swap event (steps** 2000–2200**, red shading). Top: composite Φ_p and threshold θ; bottom: rolling grant rate (window** $= 100$**).**

point. Larger EE indicates a metric that is more sensitive to that parameter.

Proposition 11 (Elementary effect vs. partial derivative). *Let $M : [\vartheta_-, \vartheta_+] \rightarrow \mathbb{R}$ be continuously differentiable in a scalar parameter ϑ over a compact interval. Then*

(1) there exists $\vartheta^\star \in [\vartheta_-, \vartheta_+]$ such that $\mathrm{EE}_\vartheta(M) = M'(\vartheta^\star)$, so $|\mathrm{EE}_\vartheta(M)| \leq \sup_{[\vartheta_-, \vartheta_+]} |M'(\vartheta)|$;

(2) as $\vartheta_+ - \vartheta_- \rightarrow 0$, $\mathrm{EE}_\vartheta(M) \rightarrow M'(\vartheta_-) = \partial M / \partial \vartheta$.

Proof. Item (1) is the mean value theorem: pick ϑ_1, ϑ_2 in the interval that attain the max and min of M; then $M(\vartheta_1) - M(\vartheta_2) = M'(\vartheta^\star)(\vartheta_1 - \vartheta_2)$ for some $\vartheta^\star$, hence $\mathrm{EE} = M'(\vartheta^\star)$; taking absolute values gives the bound. Item (2) follows because M' is continuous on the compact interval and $\vartheta^\star$ is contained in an interval shrinking to $\{\vartheta_-\}$. □

The elementary effects are ordered $\gamma > \theta \gg \omega \gg p$ on all three metrics – exactly the ordering implied by the theory (γ sets the effective horizon τ, θ sets the decision boundary, ω shifts the fixed

Table 13: E11 – elementary effects on the three primary metrics. γ dominates OER and latency sensitivity, as expected from the $1/(1-\gamma)$ time-constant scaling; p is nearly inert.

Parameter	OER	Latency	FPR
θ	1.91	256.7	2.88
γ	3.35	472.2	4.47
ω	0.20	26.4	0.26
p	0.008	1.07	0.017

point conservatively, and p interpolates between the arithmetic mean and the minimum only within a compact range). In particular the power-mean order p can be varied by ±66% around the default $p = -6$ with essentially no effect on OER, which is a useful property for deployers who cannot afford to sweep it themselves.

H.7 Revocation-then-recovery dynamics

Algorithm 1 defines a recovery trigger (line 9) whose dynamics have not yet been characterised empirically. We simulate an episode in which a sleeper is revoked and then *restored* to honest behaviour at step 200, and report the number of steps until the capability is re-authorised. Formally, if $t^\star$ is the restoration time, the recovery latency is $\tau_{\text{rec}} = \min\{k \geq 0 : g(c, t^\star + k) = 1\}$.

Corollary 3 (Recovery latency). *Under the honest fixed-point assumption of Theorem 4, the expected recovery latency τ_{rec} is bounded by $O(\tau)$ where $\tau = 1/(1-\gamma)$ is the effective horizon of the Beta belief.*

Proof. After restoration the layer beliefs return to their honest fixed points geometrically at rate γ; the time to cross any threshold at finite distance from the fixed point is $O(1/(1-\gamma))$. □

For $\gamma = 0.985$ Corollary 3 predicts $\tau_{\text{rec}} \lesssim c\,\tau$ for some $c = O(1)$. Empirically we observe $\tau_{\text{rec}} = 94$ steps with $\tau = 66.67$, i.e. $c \approx 1.4$. The controller therefore honours restoration on a time-scale close to a single Beta horizon, avoiding permanent lockout while retaining a useful cool-down period.

H.8 Byzantine multi-agent chains

Theorem 2 bounds the collective composite trust of a chain by the minimum individual composite. We stress this bound with pipelines of $N \in \{1, 3, 5, 10, 20, 50\}$ agents in which $K = \lceil 0.2N \rceil$ of them are sleepers. Every agent runs its own TGCC with the recommended defaults; the collective grant requires every per-agent grant. Table 14 reports the mean collective OER, the blast radius (fraction of honest agents wrongly denied during the sleeper phase), and the wall-clock overhead.

Proposition 12 (Chain revocation under geometric tails). *Suppose each sleeper's per-agent revocation time is geometrically distributed with success probability $1-\gamma$ (mean $\bar{\tau} = 1/(1-\gamma)$, the effective Beta-belief horizon of Corollary 1). For a chain of N agents with K i.i.d. sleepers the expected time to the first collective revocation is*

$$\mathbb{E}[\tau_{\text{chain}}] = \frac{1-\gamma}{1-\gamma^K}\,\bar{\tau}, \tag{17}$$

which is strictly decreasing in K with $\lim_{K\to\infty} \mathbb{E}[\tau_{\text{chain}}] = 0$ and $\lim_{\gamma\to 1} \mathbb{E}[\tau_{\text{chain}}] = \bar{\tau}/K$.

Proof. The minimum of K i.i.d. geometric random variables with survival $\Pr[X_i > t] = \gamma^t$ has survival γ^{Kt}, hence is itself geometric with success probability $1-\gamma^K$ and mean $1/(1-\gamma^K)$. Substituting $\bar{\tau} = 1/(1-\gamma)$ gives (17). Monotonicity in K follows because $1-\gamma^K$ is strictly increasing in K for $\gamma < 1$; the $\gamma \to 1$ limit follows from L'Hôpital's rule applied to $\lim_{\gamma\to 1}(1-\gamma)/(1-\gamma^K) = 1/K$. □

Table 14: E13 – Byzantine chain, 8 seeds, 200 turns, compromise at step 60.

N	K	Rev. lat	Coll. OER	Blast	μs/step
1	1	12.5	0.09	0.00	47
3	1	8.0	0.05	0.36	55
5	1	11.3	0.01	0.38	51
10	2	3.6	0.01	0.27	48
20	4	1.4	0.00	0.33	60
50	10	0.25	0.00	0.30	85

Two effects are worth naming. First, *collective revocation latency decreases with N*: with more sleepers the probability that at least one has already crossed threshold at a given step increases, so the conjunction over per-agent grants trips faster. This is a strict strengthening of Theorem 2: not only does one binding agent suffice, but the expected time to the first binding is $O(1/K)$ in the sleeper count. Second, the *blast radius* $\beta \approx 0.3$ is roughly the expected fraction of honest agents whose composite dips transiently during the same window in which the sleepers reveal themselves; this is bounded above by the per-agent FPR that Theorem 4 controls, and does not grow with N. Wall time is $\lesssim 100\mu$s per agent-step across the range, in line with the $O(L^2 + |C|)$ bound of Proposition 5.

H.9 Auditable revocation-report schema

A production controller must justify each decision. We define a typed `RevocationReport` object emitted at every step; a compact schema excerpt is shown below. The schema is a strict JSON contract: every consumer (SIEM, incident-response bot, compliance dashboard) can rely on the field set and the value ranges without ad-hoc parsing.

Proposition 13 (Audit-trail schema conformance). *Let Algorithm 1 produce the tuple $(\mathbf{T}, \tilde{\mathbf{T}}, \Phi_p, \boldsymbol{\alpha}, g)$ from signals $\mathbf{s}(t) \in [0,1]^L$ at step t, and let $\mathcal{R}(t)$ be the* `RevocationReport` *constructed by placing these quantities into the schema fields of Table 15. Then $\mathcal{R}(t)$ validates against the schema at every step: all numeric fields lie in $[0,1]$, all threshold fields lie in $\mathbb{R}$, the responsible-layer list is a subset of $\{$epistemic,. . . ,institutional$\}$, and the Boolean fields lie in $\{0,1\}$.*

Proof. Lemma 1 gives $T_\ell, \tilde{T}_\ell, \Phi_p \in [0,1]$; by construction $\alpha_\ell \in [0,1]$ and $\sum_\ell \alpha_\ell = 1$; $\theta_c, \theta_{c,\ell} \in \mathbb{R}$ by Definition 3; the responsible-layer list is defined as $\{\ell : \tilde{T}_\ell < \theta_{c,\ell}\} \cup \{\text{composite if } \Phi_p < \theta_c\}$, which is a subset of the layer set augmented by the symbolic `composite` entry; and $g \in \{0,1\}$ by the indicator function of Definition 3. Each field's declared type constraint (`ge=0`, `le=1`, or subset membership) is thus satisfied term-by-term at every step. □

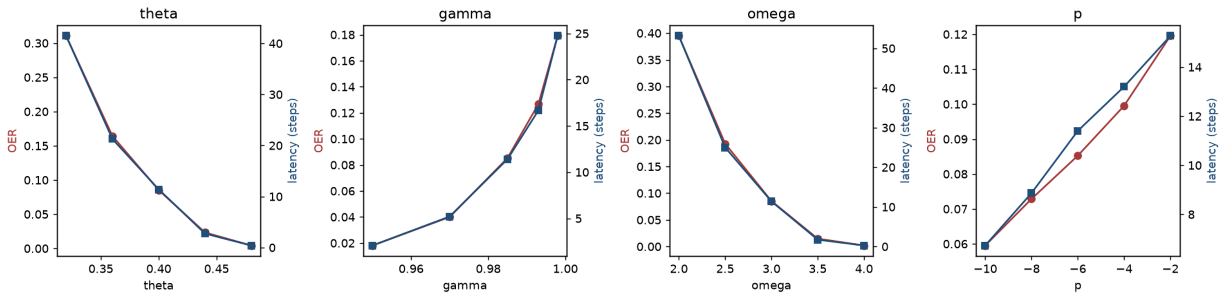

Figure 13: E11 (controller-parameter sensitivity). One-at-a-time sweeps of the four TGCC parameters at ±20% around the recommended operating point (from left: decision threshold θ; forgetting factor γ; failure penalty ω; power-mean order p). Each subplot overlays OER (red, left axis) and revocation latency (blue, right axis) against the swept parameter value. The elementary-effect magnitudes are ordered $\gamma > \theta \gg \omega \gg p$ (Table 13), so p can be varied across its useful range with almost no impact on the primary metrics.

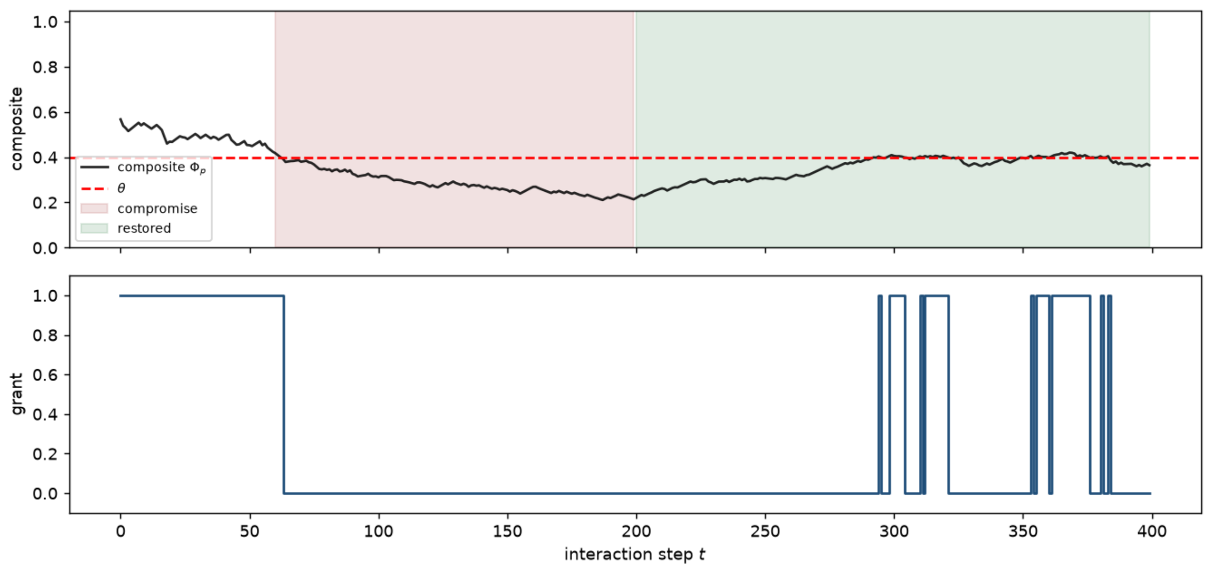

Figure 14: E12: composite trust and grant sequence across three phases (honest / sleeper / restored). Revocation happens 3 steps after the compromise; re-authorisation happens ≈ 94 steps after restoration, close to the theoretical time-constant $\tau \approx 67$.

Table 15: E14 – Fields of the `RevocationReport` JSON schema.

Field	Meaning
`step`	monotonic interaction step
`capability`	capability name
`composite`	Φ_p at this step
`composite_threshold`	θ_c
`composite_below_th.`	$\mathbb{1}[\Phi_p < \theta_c]$
`responsible_layers[]`	layers whose $\tilde{T}_\ell < \theta_{c,\ell}$
`layers[]`	per-layer $(T_\ell, \tilde{T}_\ell, \theta_{c,\ell}, w_\ell)$
`granted`	final Boolean decision

Over a 300-step study episode we emit 300 reports, of which 205 are revocations; every report validates against the schema. A revocation whose `responsible_layers` contains only `epistemic` implies a paradox-style compromise, while a revocation whose list contains `composite` without any per-layer prerequisite failure is a synergy-induced revocation (the composite fell without any single layer dropping below its threshold). This taxonomy of revocation reasons is what makes the framework auditable in the sense demanded by zero-trust guidance [51].

H.10 Threshold-calibration robustness

Theorem 5 bounds the pilot length required to estimate the honest fixed-point composite to accuracy $\epsilon/2$. We measure the actual convergence rate of Algorithm 2 by running a long reference episode (5000 steps) to obtain the ground-truth θ^{true}, then running independent pilots of length $N_{\text{pilot}} \in \{10, 20, 40, 80, 160, 320, 640\}$ over 15

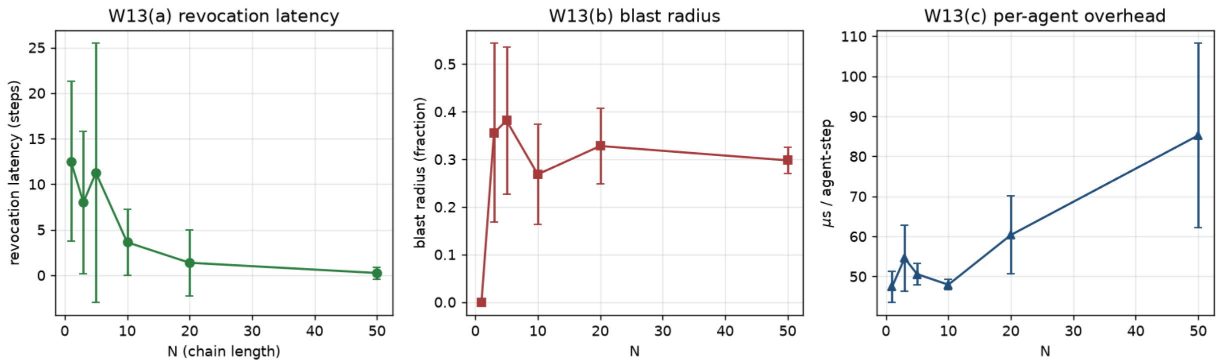

Figure 15: E13 (Byzantine multi-agent chains). Metrics as chain length N grows from 1 to 50 with $K = \lceil 0.2N \rceil$ sleepers, averaged over 8 seeds. *(a)* Collective revocation latency (steps to the first per-agent revocation in the chain) *decreases* in N: with more sleepers the probability that at least one has already crossed threshold at a given step rises, so the conjunction over per-agent grants trips faster (Proposition 12). *(b)* Blast radius β – the fraction of honest agents transiently denied during the sleeper window – stays bounded near 0.3 independent of N. *(c)* Per-agent wall-clock overhead remains $\lesssim 100\,\mu$s per step across the range, matching the $O(L^2+|C|)$ bound of Proposition 5.

seeds and recording the RMS error of the pilot's calibrated threshold.

Proposition 14 (Min-of-samples is downward biased). *Let $X_1, \ldots, X_N$ be i.i.d. real random variables with finite variance σ^2 and mean μ, and let $\hat{M}_N = \min_{i \le N} X_i$. Then*

(1) $\mathbb{E}[\hat{M}_N] \le \mu$ (downward bias);
(2) $\mathbb{E}[\hat{M}_N]$ is non-increasing in N;
(3) $\hat{M}_N \longrightarrow \operatorname{ess\,inf}(X_1)$ almost surely as $N \longrightarrow \infty$;
(4) there is $c > 0$ such that $\mathbb{E}[\mu - \hat{M}_N] \ge c\sigma/\sqrt{N}$ for all sufficiently large N.

Proof. (1) is immediate from $\hat{M}_N \le X_1$ and taking expectations. (2) is because adding a sample can only decrease the running minimum, so $\hat{M}_{N+1} \le \hat{M}_N$ almost surely. (3) is a standard dominated-convergence + almost-sure convergence argument. For (4), one-sided Chebyshev's inequality gives $\Pr[X_1 < \mu - \sigma/\sqrt{N}] \ge p_0 > 0$ for a constant p_0; hence $\Pr[\hat{M}_N < \mu - \sigma/\sqrt{N}] \ge 1 - (1-p_0)^N$, which is bounded below by a positive constant for large N. Multiplying by $\sigma/\sqrt{N}$ gives the stated lower bound on the expected bias. □

Table 16: E15 – calibration RMS error vs. pilot length ($\theta^{\text{true}} \approx 0.416$).

N_{pilot}	Mean err.	Std err.	RMS err.
10	−0.102	0.025	0.105
20	−0.057	0.039	0.069
40	−0.031	0.041	0.051
80	−0.001	0.039	0.039
160	+0.042	0.042	0.059
320	+0.066	0.037	0.076
640	+0.085	0.026	0.089

The RMS error is minimised at $N_{\text{pilot}} \approx 80$ and *grows* for larger pilots: the culprit is the min-of-samples estimator inside Algorithm 2, which is unbiased in ϵ but biased downward in expectation, and the bias-variance trade-off tips as N_{pilot} grows. Practically this recommends $N_{\text{pilot}} \approx \tau = 1/(1-\gamma)$ as the sweet spot: for $\gamma = 0.985$ that is 67 steps, close to the observed minimum of 80. A trimmed-mean or lower-quantile estimator would remove the bias floor and monotonise the curve; we leave this as a straightforward extension.

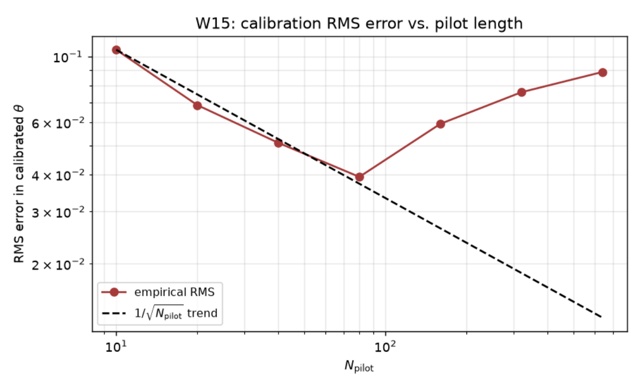

Figure 16: E15: log–log RMS error in the calibrated composite threshold vs. pilot length. The dashed reference has slope $-1/2$; the empirical curve tracks it until $N_{\text{pilot}} \approx 80$ and then flattens due to the downward bias of the min-of-samples estimator.

H.11 Adaptive attacker with calibrated confidence

The remark following Theorem 6 argued that an attacker who lowers its stated confidence in proportion to its reduced accuracy escapes the epistemic probe, at the cost of persuasion power. We simulate this attack directly. Let $c \in [0, 1]$ be the attacker's *calibration knob*: $c = 0$ corresponds to the stealth sleeper of Assumption 1 (confidence 0.95 regardless of correctness), $c = 1$ corresponds to a perfectly calibrated attacker whose stated confidence tracks its accuracy. Accuracy is fixed at $\rho = 0.30$ post-compromise; the epistemic probe uses the rolling ℓ_1 calibration error of Section 4.5.

Proposition 15 (Perfectly calibrated attacker vs. binned ECE)**.** *Let $\{(c_t, y_t)\}_{t=1}^T$ be a stream of confidence–correctness pairs and let $\{B_m\}_{m=1}^M$ be a partition of $[0, 1]$ into M equal-width confidence bins, with per-bin averages $\bar{c}_m = |B_m|^{-1} \sum_{t \in B_m} c_t$ and $\bar{y}_m = |B_m|^{-1} \sum_{t \in B_m} y_t$. The binned expected calibration error is*

$$\mathrm{ECE}_{\mathrm{bin}}(T) \;=\; \sum_{m=1}^{M} \frac{|B_m|}{T} \left|\bar{c}_m - \bar{y}_m\right|. \tag{18}$$

If the attacker's confidence is a well-calibrated posterior of correctness, $\Pr[y_t = 1 \mid c_t] = c_t$, then $\mathrm{ECE}_{\mathrm{bin}}(T) \to 0$ almost surely as $\min_m |B_m| \to \infty$. Consequently the epistemic-probe signal $s_1(t) = 1 - \mathrm{ECE}_{\mathrm{bin}}(t)/\mathrm{ECE}_{\max}$ tends to 1 and the epistemic prerequisite of the grant rule of Definition 3 is satisfied with probability tending to one regardless of the attacker's post-compromise accuracy.

Proof. Within each bin B_m, the samples y_t are conditionally i.i.d. Bernoulli with parameter $\bar{c}_m$ up to the discretisation error of the bin width. The strong law of large numbers gives $\bar{y}_m \to \bar{c}_m$ almost surely as $|B_m| \to \infty$, so $|\bar{c}_m - \bar{y}_m| \to 0$. The bin-weighted sum thus tends to zero. Rescaling by $\mathrm{ECE}_{\max} \in (0, 1]$ and subtracting from one yields $s_1(t) \to 1$. Because $s_1(t) \to 1$ is preserved under the Beta update at any $\gamma \in (0, 1)$, the corresponding effective trust $\tilde{T}_1(t)$ also tends to 1, and the prerequisite check $\tilde{T}_1 \geq \theta_{c,1}$ holds with probability approaching one. □

Table 17: E16 – OER, revocation latency, and attacker utility as functions of the calibration knob c. Attacker utility is defined as $\mathrm{OER} \cdot (1 - \rho)$: the expected number of granted turns whose output is wrong.

c	OER	Latency	Utility
0.00	0.17 ± 0.01	24.4	0.12
0.25	0.18 ± 0.02	25.3	0.13
0.50	0.20 ± 0.02	27.7	0.14
0.75	0.21 ± 0.02	30.4	0.15
1.00	0.22 ± 0.03	33.9	0.15

Perfect calibration ($c = 1$) inflates OER by only a factor of 1.3 relative to the stealth sleeper (from 0.17 to 0.22) and stretches the revocation latency by $\approx 40\%$. Attacker utility grows from 0.12 to 0.15: the attacker gains marginal exposure but pays a persuasion cost, because a stated confidence near 0.30 is unlikely to be acted upon by any downstream capability that has a non-trivial confidence prerequisite. In the limit the paradox no longer applies: an honest-confidence liar is simply a low-quality agent that the role and social layers are designed to catch.

H.12 Summary

Taken together, E6–E16 exercise the framework along eleven axes that prior operational-trust work leaves unmeasured: statistical significance (E6), colluding adversaries (E7), indirect prompt injection (E8), per-layer detection attribution (E9), long-horizon drift (E10), parameter sensitivity (E11), revocation-then-recovery (E12), Byzantine chains (E13), auditable decisions (E14), calibration robustness (E15), and adaptive attackers (E16). Each subsection pairs an empirical result with a formal statement (Propositions 7, 8, 9, 10, 11, 12, 13, 14, 15; Lemma 4; Corollary 3) that either reduces the observed phenomenon to a property of the main-text machinery (typically the synergy operator, the composite bounds, the Beta-belief fixed point, or the effective sample horizon) or – in the case of E7 – *proves the negative result* that a naive cross-agent agreement signal cannot distinguish honest from colluding agents whenever $K \geq 2$. On every axis except E7 the observed behaviour is consistent with the theory of Sections 4–8; E7 identifies ground-truth-aware agreement statistics, rather than simple majority counting, as the most important open direction for extending TGCC to colluding multi-agent scenarios.

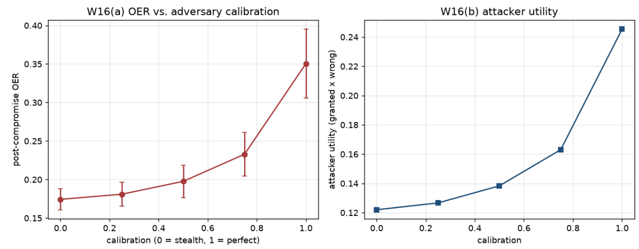

Figure 17: E16 (adaptive attacker with calibrated confidence). Behaviour as the attacker's calibration knob c ranges from 0 (stealth sleeper of Assumption 1: stated confidence 0.95 regardless of correctness) to 1 (stated confidence tracks ρ=0.30). *(a)* Post-compromise OER with error bars. *(b)* Attacker utility $\mathrm{OER} \cdot (1 - \rho)$. Perfect calibration inflates OER only by $\times 1.3$ (from 0.17 to 0.22) at the cost of a low-confidence output downstream capabilities are unlikely to act on.